

Acceptance of Ethnic and Religious Discrimination among Adolescents in Dutch-Language Belgian Secondary Schools

The Role of Intergroup Contact, Migrant Background, and
Discrimination Experiences

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1 Summary

This thesis examines how adolescents in Dutch-language Belgian secondary schools judge whether ethnic and religious discrimination is acceptable. It focuses on four aspects of their social experience: contact with people from other ethnic or religious groups, personal experiences of discrimination, witnessing discrimination against others, and migrant background.

The study uses survey data collected in 2025. The main analyses are based on 517 students and examine ethnic and religious discrimination separately. More frequent contact with people from other ethnic or religious groups was consistently associated with a lower likelihood of accepting discrimination. Adolescents who reported witnessing discrimination against others were more likely to express some acceptance of both ethnic and religious discrimination. Personal experiences of discrimination showed a more mixed pattern: perceived religious discrimination was associated with lower acceptance of religious discrimination, while perceived ethnic discrimination was not clearly associated with acceptance of ethnic discrimination. Migrant background was not clearly related to either outcome.

Because the study is cross-sectional, the findings show associations rather than cause and effect. Even so, they suggest that adolescents' views about discrimination are linked not only to their own experiences, but also to the wider social environments in which intergroup contact and discriminatory incidents occur. The findings highlight the importance of creating meaningful opportunities for contact across group boundaries while also ensuring that schools respond clearly and consistently when discrimination occurs.

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2 Introduction

Across Western European societies, increasing ethnic and religious diversity has become a defining feature of everyday social life. Despite great institutional and legal reforms to outlaw overt and institutional discrimination and promote equality, discriminatory dynamics persist in more subtle, normalized, and socially embedded forms. Rather than manifesting solely through explicit hostility or overt exclusion, present discrimination often manifests through everyday interactions, institutional practices, and moral justifications that render unequal treatment acceptable or legitimate under certain conditions (De Coninck et al., 2025b; Esses, 2021). Understanding how discriminatory behaviors' acceptance and normalization emerge remains a pressing societal and academic challenge.

Much of the existing research on discrimination has focused on either prejudicial attitudes or the direct experiences of those targeted by discrimination. These works are instrumental and essential in demonstrating the wide-ranging consequences of discriminatory treatment, especially for psychological well-being and identity development (Benner et al., 2018; Schmitt et al., 2014). Adolescents who experience discrimination are more likely to report internalizing problems such as anxiety and depression, lower self-esteem, and poorer academic functioning (Benner et al., 2018; D'hondt et al., 2024; Schmitt et al., 2014). These associations span multiple domains of adolescent adjustment and are developmentally consequential (Benner et al., 2018). While this literature provides critical insight into the harms of discrimination, it leaves comparatively underexplored an equally important dimension: whether and under what conditions discrimination is perceived as acceptable, justified, or morally permissible.

The acceptance of discrimination constitutes a distinct yet closely related phenomenon. It refers not to the experience of being discriminated against, nor to explicit prejudice toward outgroups, but to normative judgments about whether unequal treatment is legitimate in principle or in practice (Kuran & McCaffery, 2008; Nørregaard, 2026). Such judgments matter because they shape the social climates in which discrimination occurs. When discriminatory behavior is viewed as understandable or deserved, it becomes more difficult to challenge, even in contexts where formal or legal equality norms are endorsed. Recent research indicates that adolescents' orientations toward discrimination and intergroup

relations are associated with multiple aspects of their social environment. Among Belgian adolescents, discrimination experiences and patterns of social-media use are associated with different profiles of discrimination acceptance (De Coninck et al., 2025a), while peer and school norms are associated with adolescents' interethnic experiences (Tropp et al., 2016). Among Belgian adults, news-media use and trust are also associated with acceptance of discrimination (Steeleman et al., 2026). Studying acceptance thus shifts the analytical focus from individual acts to the social conditions that allow discrimination to persist.

Adolescence represents a particularly important life stage for examining these dynamics. During this period, individuals undergo significant cognitive, social, and identity-related changes that directly affect how they reason about fairness, group boundaries, and social norms (Abrams & Rutland, 2008). During this stage, cognitive and identity-related development increases young people's capacity to recognize and interpret discrimination and to reflect on how their own identities are perceived by others. As adolescents mature, they also develop a more differentiated understanding of group-based inequality, including broader processes of social privilege and disadvantage (Chan & Benner, 2026). At the same time, social relationships and group norms become increasingly salient. Peers serve as important sources of information about how members of other groups should be evaluated and treated, while school norms provide broader institutional cues about acceptable intergroup behavior (Tomova et al., 2021; Tropp et al., 2016). Adolescence is also a formative period for intergroup attitudes, while concerns about belonging and affiliation can shape young people's behavior and social decision-making (Gönültaş & Mulvey, 2019; Tomova et al., 2021). Taken together, these developmental changes make adolescence a particularly relevant period for examining how young people come to evaluate whether discriminatory treatment is acceptable or unacceptable.

Moreover, adolescents' evaluations of discrimination are not formed in a social vacuum. They are shaped by both personal experiences and everyday intergroup interactions. One factor that has received sustained scholarly attention is intergroup contact. A large body of research shows that contact between members of different ethnic or religious groups is generally associated with lower prejudice, particularly when interactions occur under favorable conditions such as equal status, cooperation, and institutional support (Pettigrew & Tropp, 2006). However, the implications of contact are not necessarily uniform across groups or social contexts. Research among ethnic minority and majority youth shows that peer and school norms can shape interethnic experiences differently depending on group status. Supportive peer norms appear especially relevant to majority-group youths' immediate

comfort in cross-ethnic interactions, whereas minority-group youth may be particularly attentive to institutional signals of inclusion and exclusion within the school environment (Tropp et al., 2016). Moreover, perceived discrimination can undermine adolescents' comfort and interest in cross-ethnic interaction over time, suggesting that the benefits of contact depend partly on the broader relational climate in which it occurs (Tropp et al., 2016). Importantly, recent research among Belgian adults also indicates that more frequent contact with ethno-religious minorities is associated with lower acceptance of discrimination against these groups (Steeman et al., 2026). Together, these findings suggest that intergroup contact may be relevant not only for attitudes toward outgroups, but also for judgments about whether discriminatory treatment is acceptable.

Perceived discrimination represents a second process through which adolescents' social experiences may become relevant to such normative judgments. Experiences of discrimination communicate rejection and unequal social standing and are consistently associated with poorer adjustment during adolescence (Benner et al., 2018). They may also alter how individuals understand their position within intergroup relations. Longitudinal research suggests that perceived discrimination can increase the centrality of minority-group membership, while other work indicates that sustained experiences of rejection can contribute to disengagement from the broader national community (Jasinskaja-Lahti et al., 2009; Ramos et al., 2012). These findings indicate that discrimination is not only an adverse interpersonal experience but may also shape how individuals interpret group boundaries, belonging, and their relationship to the wider social environment. Yet its implications for evaluations of discrimination itself remain less clear. Experiencing discrimination does not necessarily translate into greater rejection of discriminatory practices. In a recent Belgian adult sample, greater personal experience with discrimination was unexpectedly associated with greater acceptance of discrimination, although the cross-sectional design does not allow conclusions about causality (Steeman et al., 2026). Thus, rather than assuming that discrimination uniformly heightens sensitivity to injustice, it may be more appropriate to consider its relationship with normative judgments as an open empirical question. Little is currently known about whether adolescents who perceive more discrimination are more or less likely to regard ethnic and religious discrimination as acceptable.

A third and comparatively understudied form of exposure is vicarious discrimination, defined as witnessing or learning about discriminatory treatment directed at others (Louie & Upenieks, 2022). Such experiences appear to be common in adolescents' everyday environments. In a representative sample of Spanish adolescents, 66% reported witnessing

discrimination, with peers identified as the most common perpetrators and schools as the primary context in which these experiences occurred (Valdivieso-León et al., 2024). Similarly, Chan and Benner (2026) found that more than 70% of adolescents reported at least one incident of vicarious discrimination over a 14-day period, including experiences involving parents, friends, strangers, and media. Although adolescents are not personally targeted in these situations, vicarious discrimination can still shape how they perceive their social environment. Research has linked witnessing discrimination to emotional distress, heightened perceptions of threat, and greater awareness of group-based inequality (Chan & Benner, 2026; Li Verdugo et al., 2024; Louie & Upenieks, 2022). Observational research further suggests that the social responses surrounding discriminatory incidents can influence how observers interpret both the incident and discrimination more broadly (Sanchez et al., 2024). Vicarious discrimination may therefore provide adolescents with information about prevailing intergroup relations and social norms, although its relationship with the acceptance of discrimination remains largely unexplored.

Taken together, existing research suggests that adolescents' evaluations of discrimination are shaped by several related aspects of their social experience, including intergroup contact, personal and vicarious discrimination experiences, and their position within intergroup relations. Intergroup contact is generally associated with more positive outgroup attitudes, but its implications can vary depending on the quality of interaction, perceived inequality, and the surrounding peer and school climate (Pettigrew & Tropp, 2006; Tropp et al., 2016). Likewise, experiences of discrimination can shape perceptions of threat, fairness, belonging, and group boundaries (Benner et al., 2018; Chan & Benner, 2026; Louie & Upenieks, 2022). These processes may also differ between majority- and minority-background adolescents, who encounter intergroup relations from different social positions (Munniksma et al., 2015; Tropp et al., 2016). Yet relatively little research has examined how these factors jointly relate to adolescents' normative acceptance of ethnic and religious discrimination.

The present study addresses this gap by examining their associations within a single empirical framework: how migrant background, intergroup contact, perceived discrimination, and vicarious discrimination are associated with adolescents' acceptance of ethnic and religious discrimination. By focusing on acceptance rather than solely on prejudice or the consequences of victimization, the study shifts attention toward the normative evaluation of discriminatory behavior itself. In doing so, it aims to clarify how adolescents' social

experiences and group position are related to whether discrimination is rejected, tolerated, or regarded as acceptable.

Research Question

How are migrant background, discrimination experiences, and intergroup contact associated with adolescents' acceptance of ethnic and religious discrimination?

3 Literature Review and Theoretical Framework

3.1 Prejudice and discrimination

To clarify the conceptual scope of the present study, it is important to clearly distinguish between two key and overlapping concepts: prejudice and discrimination. A classic definition of prejudice by Allport (1954, p. 9) is that “prejudice is an antipathy based on a faulty and inflexible generalization. It may be felt or expressed. It may be directed toward a group as a whole or toward an individual because he is a group member.” More recent work similarly conceptualizes prejudice as involving negative attitudes and affective evaluations directed toward social groups (Esses, 2021). Discrimination, on the other hand, is commonly understood as the unequal or less favorable treatment of individuals based on their actual or perceived membership in a social group, such as ethnicity, religion, gender, or other socially salient characteristics (Brown & Bigler, 2005; Esses, 2021). While discrimination is often discussed alongside prejudice, the two concepts are analytically distinct. In simple terms, prejudice is internal and affective; a belief, while discrimination is external and behavioral; an action. Discrimination in this case can be understood as the active expression of prejudice. However, individuals may reject prejudicial attitudes while simultaneously accepting certain forms of unequal treatment as legitimate or necessary. This distinction is particularly relevant in contemporary societies, where overt expressions of prejudice are increasingly stigmatized, yet more subtle or conditional forms of exclusion persist (De Coninck et al., 2025b; Nørregaard, 2026).

However, this study focuses not on discriminatory attitudes nor prejudicial beliefs, but specifically on the acceptance of discrimination, which is the extent to which individuals

perceive unequal treatment of certain groups as justified, understandable, or morally permissible. Acceptance of discrimination differs from personal discriminatory intent and from direct experiences of being discriminated against. Rather, it reflects a normative evaluation of social practices and institutional arrangements. Such evaluations are shaped not only by personal beliefs but also by dominant cultural narratives, political discourse, and observed social norms (Kuran & McCaffery, 2008; Nørregaard, 2026).

Empirical research shows that acceptance of discrimination is multidimensional and varies systematically across different grounds such as ethnicity, religion, gender identity, sexual orientation, age, and disability. For example, De Coninck et al. (2025a) found that among Belgian adolescents, there are distinct profiles of discrimination acceptance, ranging from broadly egalitarian orientations to selective and generalized acceptance of unequal treatment across multiple domains. Importantly, these profiles indicate that individuals may reject discrimination in some domains while condoning it in others, particularly in relation to gender and sexuality. Similar domain-specific variation has been documented in adult populations, where acceptance levels differ substantially depending on the target group, with higher tolerance for discrimination against socially stigmatized or marginalized groups (De Coninck et al., 2025a; Kuran & McCaffery, 2008). These findings show the multilayered nature of the acceptance and tolerance of discrimination and how it is domain-dependent.

Moreover, the literature shows that contextual justifications strongly shape moral evaluations of discriminatory practices. Experimental evidence shows that individuals are more likely to perceive discrimination as morally acceptable when it is justified by reference to statistically accurate group differences, even when the outcome entails unequal treatment (Nørregaard, 2026). By contrast, when discriminatory decisions are framed as based on inaccurate or biased information, moral condemnation increases substantially. Notably, the perceived intention of the discriminator appears to play a comparatively limited role in shaping moral judgments, as respondents focus primarily on the epistemic validity of the justification provided (Nørregaard, 2026). This suggests that appeals to rationality and efficiency can legitimize discriminatory outcomes, complicating normative commitments to formal equality.

Beyond explicit moral evaluations, acceptance of discrimination is also reflected in subtle, everyday practices of social distance. Using a conjoint experimental design, De Coninck, Steeman, and d’Haenens (2025b) demonstrate systematic preferences to avoid proximity to certain groups in mundane contexts such as public transport seating choices. Strong negative biases were observed toward Muslims, people of Moroccan and Nigerian

origin, transgender individuals, and people with stigmatized physical characteristics such as being overweight or having visible piercings. These patterns point to the persistence of covert discrimination in everyday interactions, even in contexts where overt discriminatory behavior is socially sanctioned. Such findings highlight that acceptance of discrimination is not only articulated through explicit policy preferences or moral reasoning but is also enacted through daily micro-level choices.

Individual-level characteristics further structure variation in acceptance of discrimination. Gender differences are consistently observed, with men reporting higher acceptance of discriminatory practices across multiple domains compared to women (De Coninck et al., 2025a; Kuran & McCaffery, 2008). However, within-gender heterogeneity is substantial, particularly among adolescents, where multiple attitudinal profiles coexist (De Coninck et al., 2025a). Moreover, experiences of discrimination are associated with more exclusionary orientations in some groups, indicating that lived experiences of unfair treatment do not uniformly translate into solidarity with other marginalized groups but may also foster defensive boundary drawing (De Coninck et al., 2025a; Steeman et al., 2026).

Finally, media environments play a structuring role in shaping normative boundaries around discrimination. Evidence from Belgian adolescents indicates that both the intensity and platform-specific patterns of social media use are associated with differential profiles of discrimination acceptance (De Coninck et al., 2025a). Certain platforms are linked to more egalitarian orientations, while others are associated with a higher likelihood of exclusionary profiles, particularly among boys. These findings suggest that acceptance of discrimination is embedded in mediated social contexts, where platform-specific norms, exposure to exclusionary discourse, and algorithmic curation can normalize or contest discriminatory attitudes.

Taken together, the literature conceptualizes acceptance of discrimination as a context-dependent, multidimensional phenomenon shaped by justificatory frames, social categorization processes, everyday practices of social distance, individual characteristics, and media environments.

3.2 Ethnic and Religious Discrimination

Ethnic and religious discrimination are closely related, but they should not be treated as identical. Ethnic discrimination generally refers to unequal treatment based on perceived

origin, nationality, ethnicity, race, or skin colour, whereas religious discrimination concerns unequal treatment based on religious affiliation, identity, practice, or visibility. This distinction matters because prejudice, stereotyping, and discrimination can operate through different social categories and at different levels, including individual interactions, institutional practices, and broader cultural processes (Dovidio et al., 2010). In the context of adolescence, where young people are actively developing social identities and moral evaluations of fairness, these categories can become meaningful bases for both belonging and exclusion (Benner et al., 2018; Van der Straten Waillet & Roskam, 2012).

In Western European societies, ethnicity, migration background, and religion often overlap socially, particularly for adolescents from migrant-background families. However, overlap does not imply equivalence. Van der Bracht et al. (2016), studying adolescents in Flemish secondary schools, note that concerns about immigrant religiosity, especially Muslim religiosity, have become increasingly salient in European public debates and school contexts. Their findings show that ethnic school composition is associated with adolescents' religious salience: pupils in schools with a higher share of ethnic minority students tend to report higher levels of religious salience, and this association differs by migrant generation and religious affiliation (Van der Bracht et al., 2016). These findings suggest that schools do not merely reflect pre-existing religious identities; they can also shape when and how religious identity becomes socially salient.

This is particularly relevant in Belgium, where religious identity is embedded in a specific institutional and educational context. Van Praag et al. (2016) argue that, although Flemish society contains historically confessional institutions, Islam does not occupy an equivalent institutional position within education and civil society. They also note that Islamic religiosity is often treated as controversial in relation to immigrant minorities' educational incorporation. Their qualitative study shows that religion is sometimes interpreted through migration and ethnicity, but also that Muslim students may experience religiosity as a source of belonging, motivation, and moral orientation. At the same time, religious visibility can expose students to exclusion, for instance when wearing a headscarf becomes associated with discrimination or peer exclusion (Van Praag et al., 2016). This supports the idea that religion is not simply a private belief system, but can become a visible and socially evaluated marker in school life.

Religious discrimination also deserves separate analytical attention because religious intergroup relations during childhood and adolescence remain less extensively studied than ethnic, racial, or gender-based intergroup relations. Van der Straten Waillet and Roskam

(2012) argue that religious identity differs from many other social categories because it is connected not only to group membership, but also to beliefs, values, practices, and moral commitments. Their study conceptualizes religious discrimination as linked to both prejudice and moral reasoning, and finds that religious discrimination varies by age, school environment, group status, and parental messages about other religious groups (Van der Straten Waillet & Roskam, 2012). This is directly relevant to the present study because acceptance of discrimination is also a normative judgment: it concerns whether unequal treatment is viewed as fair, justified, or permissible.

At the same time, ethnic discrimination remains a central dimension of adolescents' social experiences. Benner et al. (2018), in a meta-analysis of racial and ethnic discrimination during adolescence, show that perceived racial or ethnic discrimination is associated with multiple negative outcomes, including socioemotional distress, lower self-esteem, weaker academic engagement, and greater involvement in risky behaviors. Schmitt et al. (2014) similarly show that perceived discrimination is negatively related to psychological well-being across different stigmatized groups, with particularly important implications for disadvantaged groups and younger populations. These findings demonstrate that ethnic discrimination is not merely an abstract form of bias, but a developmentally consequential experience that can shape adolescents' well-being, identity, and interpretation of social relations (Benner et al., 2018; Schmitt et al., 2014).

Belgian school-based research further illustrates how ethnic discrimination operates in adolescents' everyday educational environments. D'hondt et al. (2024) show that ethnic discrimination can take subtle forms, including ethnic microaggressions, and emphasize that microaggressions are not interchangeable with more blatant forms of discrimination. Their study of students of immigrant descent in Belgian secondary schools considers Muslim affiliation, country of origin, and generational status, showing that ethnicity, religion, and migration background can intersect in shaping minority students' experiences of exclusion (D'hondt et al., 2024). Similarly, D'hondt et al. (2017) show that ethnic teacher discrimination among Turkish and Moroccan immigrant adolescents in Belgium is related to school misconduct and is embedded in broader processes of ethnic and host-national identification. These findings support treating ethnicity and religion as connected, but multidimensional, bases of social positioning in school contexts.

The complexity of the ethnic-religious relationship is also visible in research on Muslim adolescents. Wenzing et al. (2024) show that, for Muslim minority adolescents, perceived ethnic and religious discrimination may be difficult to separate empirically, as

ethnic and religious discrimination items loaded onto a joint perceived ethno-religious discrimination construct. However, the same study also finds that ethnic and religious identification remain distinct dimensions of identity. In other words, adolescents may experience discrimination as ethno-religiously entangled while still distinguishing between ethnicity and religion as aspects of who they are. This finding is especially important for the present study because it suggests that ethnic and religious discrimination are neither fully separate nor fully identical. Rather, they are overlapping domains that may operate differently depending on whether one examines identity, contact, discrimination experience, or acceptance of unequal treatment (Wenzling et al., 2024).

For this reason, the relationship between ethnic and religious discrimination is best understood as one of overlap rather than equivalence. For many migrant-background adolescents, ethnic and religious identities may be socially intertwined, especially when religion is interpreted through assumptions about migration, origin, culture, or belonging. However, religious discrimination may also operate through distinct mechanisms when religious practices, symbols, or affiliations become visible in school life or subject to peer judgment and institutional regulation (Van der Bracht et al., 2016; Van Praag et al., 2016). Conversely, ethnic discrimination may occur without explicit reference to religion, through assumptions about nationality, language, skin colour, origin, or immigrant status (Benner et al., 2018; D'hondt et al., 2017). These differences matter because adolescents may encounter ethnic and religious outgroups differently, experience ethnic and religious discrimination differently, and evaluate the acceptability of ethnic and religious discrimination in different ways.

This domain distinction is central to the present study. Rather than assuming that ethnic and religious discrimination form a single interchangeable construct, this study treats them as related but analytically distinct domains. For the ethnic domain, the analysis examines ethnic intergroup contact, perceived ethnic discrimination, and acceptance of ethnic discrimination. For the religious domain, it examines religious intergroup contact, perceived religious discrimination, and acceptance of religious discrimination. This domain-matching strategy allows the study to assess whether similar theoretical mechanisms operate across ethnic and religious forms of discrimination, while avoiding the assumption that contact, discrimination experiences, and acceptance function identically across both domains.

3.3 Adolescence

This study focuses on adolescence, which constitutes a particularly important developmental phase for examining these normative judgments. During this period, individuals develop more advanced cognitive capacities for moral reasoning, perspective-taking, and abstract thinking, which allow them to reflect on fairness, legitimacy, and social rules beyond immediate personal experience (Abrams & Rutland, 2008). This period constitutes a critical developmental period for the formation of social norms, moral reasoning, and intergroup attitudes (Abrams & Rutland, 2008; Brown & Bigler, 2005; Gönültaş & Mulvey, 2019; Tropp et al., 2016). As a result, adolescents become increasingly capable of evaluating whether unequal treatment of others is acceptable, rather than merely reacting emotionally to instances of unfairness. These evaluations are not formed in isolation but are embedded within peer cultures, institutional settings, and broader societal debates that adolescents actively engage with.

From a social-developmental perspective, adolescence is a period in which group boundaries and social identities become increasingly salient (Abrams & Rutland, 2008). The development of subjective group dynamics implies that adolescents are particularly attuned to group norms, ingroup loyalty, and the social regulation of deviance within peer groups (Abrams & Rutland, 2008). As peer groups gain importance relative to family during this period, adolescents become more sensitive to perceived peer norms regarding inclusion, exclusion, and the legitimacy of unequal treatment. This heightened normative sensitivity implies that attitudes toward discrimination are not merely individual moral positions, but are embedded within peer-based normative climates (Tropp et al., 2016).

At the same time, adolescence is marked by an intensified orientation toward social belonging and the avoidance of social exclusion. Neurodevelopmental and social-cognitive research shows that adolescents display heightened responsiveness to peer evaluation and social acceptance, rendering them particularly susceptible to normative pressures (Tomova et al., 2021). This sensitivity to social risk implies that adolescents may align their expressed attitudes with perceived dominant norms within their peer environments, including norms concerning intergroup relations and the acceptability of discriminatory practices. Consequently, acceptance of discrimination during adolescence may reflect not only internalized beliefs, but also strategic conformity to peer-based normative expectations.

Finally, schools function as central socialization contexts during adolescence, structuring daily intergroup contact, peer norm formation, and exposure to institutional

authority. School environments provide repeated opportunities for adolescents to observe how diversity, inclusion, and inequality are handled in practice, thereby shaping their normative expectations regarding the legitimacy of differential treatment (Munniksma et al., 2015; Tropp et al., 2016; Van der Bracht et al., 2016). In contexts characterized by ethnic or religious segregation, the social boundaries between groups may be reinforced, whereas more heterogeneous school environments may foster normative climates in which exclusionary practices are more readily questioned (Tropp et al., 2016; Van der Bracht et al., 2016).

Taken together, adolescence represents a formative period in which normative orientations toward discrimination are actively constructed through the interplay of cognitive development, identity formation, peer norm sensitivity, and institutional socialization. This makes adolescence a theoretically and empirically appropriate focus for examining how acceptance of discrimination emerges and how it is shaped by intergroup contact and social positioning related to outgroup members.

3.4 Intergroup Contact

3.4.1 Classical Intergroup Contact Theory and Its Core Mechanisms

Intergroup contact theory is one of the most influential frameworks for explaining how everyday interactions between members of different social groups shape intergroup attitudes. Intergroup contact theory provides the primary theoretical framework for the present study. The theory's foundational claim is that contact, under certain conditions, reduces prejudice and improves intergroup relations. Allport (1954) first introduced the theory, noting that intergroup contact helps reduce prejudice, stereotyping, and conflict. However, Allport (1954) identified four conditions under which intergroup contact is optimized: 1. equal status within the contact, which means that within an interaction, there should be no hierarchies, 2. common goals, 3. intergroup cooperation, and 4. support of authority, law, or custom. Allport's theory has laid the foundation for research on the effectiveness of contact and the conditions under which it succeeds.

Pettigrew and Tropp's (2006) meta-analysis provides strong evidence for the general effectiveness of contact across contexts, populations, and target groups, demonstrating that contact is associated with lower negative attitudes, i.e., prejudice and discrimination, and

more positive outgroup evaluations. The core theoretical premise is not simply that exposure to diversity changes attitudes, but that interpersonal interaction can reduce perceived threat, increase empathy, and facilitate the development of more inclusive social norms.

For adolescents specifically, contact is especially consequential because peer relations and social belonging are central developmental concerns. During adolescence, group boundaries become socially meaningful, and peer networks become key sites through which norms are learned and enforced (Abrams & Rutland, 2008; Tropp et al., 2016). Contact, therefore, shapes attitudes not only through individual-level psychological mechanisms but also through peer-based processes.

Importantly, the contact hypothesis has been refined to emphasize the conditions under which contact produces positive effects. Although Allport's original conditions are not always explicitly tested in survey-based research, later work demonstrates that the presence of equal status, shared goals, and institutional support generally strengthens the positive association between contact and intergroup attitudes (Pettigrew & Tropp, 2006). In adolescence, institutional contexts like schooling matter because they structure opportunities for interaction, the stability of peer relationships, the authority of teachers and the school system, and the normative environment in which those relationships unfold.

Moreover, most contact research focuses on prejudice or outgroup attitudes; these processes may also be relevant to judgments about discriminatory treatment. Greater familiarity and social proximity can make members of other groups less abstract and may therefore make unequal treatment toward them more difficult to justify. Consistent with this broader reasoning, Neumann and Moy (2018) found that several forms of interracial and interethnic contact, particularly positive interactions and cross-group friendships, were associated with more inclusive attitudes toward immigrants across European countries. The present study therefore extends the logic of intergroup contact theory from evaluations of outgroups to adolescents' normative evaluations of discrimination against those groups.

3.4.2 Contact as Interaction, Not Exposure: Quantity, Quality, and Friendship

More recent scholarship increasingly distinguishes between the quantity of contact and the quality of contact. Adolescents may attend diverse schools and still experience segregation through friendship homophily or peer group boundaries (Munniksma et al., 2015;

Tropp et al., 2016). Consequently, simple exposure to diversity is insufficient; the character of interactions matters for whether contact produces inclusive outcomes.

A particularly important form of contact is cross-group friendship. Longitudinal evidence among Belgian adolescents suggests that intergroup friendships and educational experiences can shape the development of ethnocentrism over time, highlighting that contact processes are not static but unfold across adolescence (Hooghe et al., 2013). Friendship-based contact is theorized to be especially influential because it involves repeated interaction and opportunities for perspective-taking and closeness. These features make friendships more likely than superficial contact to generate sustained attitudinal change (Hooghe et al., 2013; Munniksma et al., 2015; Tropp et al., 2016).

Research also emphasizes that contact includes both positive and negative encounters. A longitudinal study among adolescents in the Netherlands found that increases in the share of interethnic friends were associated with lower prejudice, whereas gaining interethnic foes was associated with higher prejudice. However, the study found no evidence that interethnic friendships buffered the prejudice-increasing association of negative contact (Beer ten Berge et al., 2017). This suggests that positive and negative contact can coexist and may contribute independently to adolescents' intergroup attitudes.

Moreover, in diverse peer environments, adolescents can thus experience “mixed contact ecologies,” where positive interactions coexist with experiences of rejection or exclusion (Beer ten Berge et al., 2017). These mixed ecologies are particularly relevant for explaining normative judgments about discrimination. Adolescents who experience low-quality contact may come to see discrimination as a normal or unavoidable feature of intergroup relations, whereas those who develop meaningful cross-group friendships may interpret discrimination as illegitimate or morally unacceptable.

3.4.3 Contact in Adolescence as a Normative and Group-Dynamic Process

Intergroup contact should also be understood as a normative process embedded in group dynamics rather than purely as an individual-level phenomenon. Adolescents do not form intergroup attitudes in isolation; they do so in relation to peer norms and perceived group expectations. Not only that, but also the social risks of deviating from one's in-group norms (Abrams & Rutland, 2008; Tomova et al., 2021). Peer groups can enforce boundary maintenance through exclusion and sanctioning, meaning that contact is often shaped by the

social costs and benefits attached to cross-group interaction (Abrams & Rutland, 2008; Tomova et al., 2021; Tropp et al., 2016).

Tropp et al. (2016) provide an important framework for understanding this dynamic by showing that school norms, peer norms, and discrimination jointly predict adolescents' interethnic experiences. This implies that contact is partly a function of the normative climate: in environments where peer norms endorse exclusion or where discrimination is tolerated, contact may be limited or negative. Conversely, in settings where inclusive norms are stronger, contact is more likely to be positive and to yield the classic prejudice-reducing effects identified in meta-analytic work (Pettigrew & Tropp, 2006).

This broader normative role of contact is also evident outside face-to-face settings. Imperato et al. (2021), for example, found that online intercultural contact was associated with stronger commitment to a shared online community, which in turn was related to greater recognition of racial and ethnic discrimination and anti-racist behavior. While their adult Facebook sample differs from the adolescents examined here, the findings illustrate how contact may operate partly through perceptions of shared belonging and the wider social environment rather than through interpersonal liking alone.

Because acceptance of discrimination is itself a normative judgment, these processes are directly relevant to the present study. If frequent contact reduces social distance and strengthens familiarity across ethnic and religious boundaries, adolescents with greater contact should be less likely to regard unequal treatment of those groups as acceptable. This reasoning provides the basis for expecting a negative association between domain-specific intergroup contact and acceptance of ethnic and religious discrimination.

3.4.4 Majority–Minority Asymmetries in Contact Effects

A major refinement in contact scholarship concerns the recognition that contact effects are not symmetrical for majority and minority groups. Minority adolescents often experience contact in contexts where status is unequal and where discrimination is more likely, which can fundamentally change the psychological meaning of contact. Tropp et al. (2016) explicitly demonstrate that discrimination and intergroup experiences act differently for minority versus majority youth. This aligns with a broader argument that contact may yield more reliable benefits for majority youth, while minority youth may experience contact as more ambivalent

due to vulnerability to rejection or unequal treatment. This ties back to Allport's fundamental claim that having equal status is crucial to the positive outcome of intergroup contact.

Munniksma et al. (2015) show how identity processes are involved in these dynamics by examining friendships and outgroup attitudes among ethnic minority youth and highlighting the mediating role of ethnic and host-society identification. This is theoretically important for our model because it suggests that contact can shape attitudes partly through identity-related pathways, which are not equivalent across groups. For minority youth, positive contact may encourage host-society identification and trust, whereas negative or discriminatory contact may reinforce in-group boundaries or weaken identification with the broader society.

These asymmetries help explain why migrant background is not merely a control variable but a theoretically meaningful moderator. It moderates both the typical quality of contact and the mechanisms through which contact is interpreted. When contact occurs under unequal or discriminatory conditions, its ability to reduce acceptance of discrimination may be weakened, and in some cases, contact may even coincide with increased endorsement of exclusionary norms.

3.5 Discrimination Experiences and Intergroup Contact

The final element of the framework concerns discrimination as part of the social context in which intergroup contact is experienced and interpreted. As discussed above, contact is generally associated with more positive intergroup attitudes, but its implications depend on the conditions surrounding the interaction. Contact that occurs under conditions of equality and institutional support is more conducive to positive intergroup outcomes, while perceptions of hierarchy, threat, and injustice can alter how contact relates to attitudes and behavior (Cocco et al., 2024; Pettigrew & Tropp, 2006). Similarly, communication during contact appears to matter: among advantaged-group members, contact was more strongly connected to support for social change when interactions included discussion of discrimination and differences in power rather than cultural differences alone (Tropp et al., 2021). These findings suggest that contact cannot be understood independently of the wider intergroup environment in which it occurs.

3.5.1 *Vicarious Discrimination*

One feature of this environment is vicarious discrimination, referring to secondhand exposure to discriminatory treatment directed at others, whether through witnessing an incident, hearing about another person's experience, or encountering discrimination through media (Louie & Upenieks, 2022). Such exposure is particularly relevant during adolescence. As young people develop greater capacity to recognize group-based inequality and systemic discrimination, they also become increasingly able to interpret and generalize social information obtained through observing others (Brown & Bigler, 2005; Chan & Benner, 2026). Vicarious discrimination is also common in adolescents' everyday lives, occurring through peers, family members, strangers, and especially media environments (Chan & Benner, 2026; Valdivieso-León et al., 2024). Chan and Benner's (2026) daily-diary study, for example, found that more than 70% of adolescents encountered vicarious discrimination during a 14-day period, with repeated exposure involving parents, friends, and media associated with greater negative mood, perceived safety stress, and social misfit.

Importantly, the consequences of vicarious discrimination are not limited to distress. Studies have associated vicarious discrimination with depressive symptoms, anxiety, anger, and anticipatory concerns about becoming a target oneself (Dhanani et al., 2022; Li Verdugo et al., 2024; Louie & Upenieks, 2022). Louie and Upenieks (2022), for instance, found that discrimination experienced by family members was associated with depressive and anxiety symptoms at levels comparable to direct discrimination, illustrating how discriminatory stress can extend beyond its immediate target. At the same time, witnessing discrimination can also heighten recognition of inequality. Among White Americans, Uluğ and Tropp (2021) found that witnessing anti-Black discrimination was associated with greater awareness of racial privilege and stronger willingness to engage in racial-justice action. Additionally, among adolescents, Heard-Garris et al. (2021) identified both helplessness and activism as responses to media-based vicarious racism, demonstrating that distress and resistance can coexist. More broadly, Cocco et al. (2024) conclude that awareness of injustice can mobilize responses to inequality, whereas perceived hierarchy and threat can produce different or even opposing consequences. Thus, the existing literature does not provide a strong basis for predicting that vicarious discrimination should, by itself, either increase or decrease acceptance of discrimination. Its direct association with acceptance is therefore more appropriately treated as exploratory.

The rationale for moderation, however, does not require a uniform attitudinal response. Rather, it rests on the information vicarious discrimination provides about the broader intergroup environment. Intergroup contact gives adolescents information about relations between groups through their own interpersonal encounters. Witnessing discrimination provides an additional source of information by making unequal treatment visible beyond those encounters. Uluğ and Tropp (2021), for example, argue that witnessing discrimination can reveal inequalities that advantaged-group members may otherwise have little opportunity to encounter personally, thereby shifting their understanding of racism from an exclusively interpersonal issue toward recognition of broader inequality.

The meaning drawn from witnessed discrimination also depends on the social responses surrounding it. Sanchez et al. (2024) experimentally demonstrated that observers who saw racial-discrimination disclosures met with denial subsequently responded less favorably to the person reporting discrimination, expressed less support for discussing discrimination, judged the alleged perpetrator as less racist, and were more likely to reproduce denial themselves. This shows that observing discriminatory events and the reactions surrounding them can shape how individuals interpret discrimination and the norms governing appropriate responses to it. In this sense, vicarious discrimination can provide contextual information about whether unequal treatment remains socially consequential and how seriously such treatment is taken.

This contextual process may be especially important during adolescence because peers and schools are salient sources of information about acceptable intergroup behavior. Tropp et al. (2016) found that peer and school norms predicted adolescents' interethnic experiences, while perceived discrimination altered some of these relationships. Among Latino students, supportive school norms predicted greater comfort and contact quality over time, particularly when perceived discrimination was lower. The authors suggest that when discrimination is more salient, messages supporting cross-ethnic relations may conflict with students' social reality. Although Tropp et al. examined personally perceived rather than vicarious discrimination, their findings establish a broader principle relevant here: favorable signals about intergroup relations may carry different implications when they coexist with evidence of unequal treatment.

Applied to the present study, frequent intergroup contact and vicarious discrimination may therefore provide adolescents with partly conflicting information. Greater contact may reduce social distance and provide repeated experiences across ethnic or religious group boundaries, supporting lower acceptance of unequal treatment. At the same time, witnessing

discrimination demonstrates that such interactions coexist with situations in which group membership continues to shape how people are treated. Vicarious discrimination need not negate the benefits associated with contact, nor does it necessarily make discrimination more acceptable. Rather, it may reduce the extent to which frequent intergroup contact can be interpreted as evidence of equal and inclusive intergroup relations. Accordingly, the negative association between intergroup contact and acceptance of discrimination is expected to be weaker among adolescents who report vicarious discrimination than among those who do not.

3.5.2 *Perceived Discrimination*

Beyond indirect exposure, adolescents may also experience discrimination personally. Perceived discrimination refers to the subjective recognition that one has been treated unfairly because of a socially meaningful group membership (Brown & Bigler, 2005; Schmitt et al., 2014). This distinction between discriminatory treatment and its perception is important because identifying an experience as discrimination requires both recognizing the treatment as unfair and attributing it to group membership. Schmitt et al. (2014) similarly distinguish objective encounters with unequal treatment from the subjective perception of oneself as a target, arguing that the latter carries implications for how individuals understand their position within society. Adolescence is particularly relevant to these appraisal processes because young people become increasingly able to recognize subtle forms of discrimination and understand the implications of ethnicity, race, and group membership for everyday social experience (Benner et al., 2018; Brown & Bigler, 2005).

Perceived discrimination is also socially patterned rather than randomly distributed. Migration history is strongly associated with reported discrimination (Klatzka et al., 2024). Additionally, research among ethnic-minority adolescents documents discriminatory treatment across schools and other everyday settings (Benner et al., 2018; D'hondt et al., 2017, 2024). Ethnic and religious discrimination may also overlap in practice. Among Muslim minority adolescents, Wenzing et al. (2024) found that perceived ethnic and religious discrimination were difficult to separate empirically, even though ethnic and religious identification remained distinct. This suggests that ethnicity and religion can constitute related but non-equivalent bases on which adolescents interpret unequal treatment. For the present study, this makes it important to examine perceived discrimination in relation to the corresponding domain of intergroup contact and discrimination acceptance.

A large body of research demonstrates that perceived discrimination has consequences for adolescents' psychological and social adjustment. Benner et al. (2018) found consistent associations between racial and ethnic discrimination and socioemotional distress, lower self-esteem, poorer academic adjustment, and risky behavior, while Schmitt et al. (2014) similarly found a robust negative relationship between perceived discrimination and psychological well-being across stigmatized groups. Importantly, the consequences of discrimination appear especially pronounced when unequal treatment is understood as pervasive rather than isolated. In their study, Schmitt et al. (2014) found experimentally that perceptions of widespread discrimination were more harmful than attributing a single negative event to discrimination, suggesting that perceived discrimination can convey information about an individual's broader position within the social environment.

These experiences may also affect how individuals understand their relationship with their own group and with the wider society. The rejection-identification tradition proposes that discrimination can make minority-group membership more salient, although evidence suggests that these identity responses are more differentiated than a simple increase in ingroup identification. Ramos et al. (2012), for example, found longitudinally that perceived discrimination predicted greater subsequent centrality of minority-group membership, rather than uniformly strengthening all dimensions of group identification. Jasinskaja-Lahti et al. (2009) found a different response: perceived ethnic discrimination predicted lower identification with the Finnish national community, which in turn was associated with more negative attitudes toward the national outgroup. Together, these studies suggest that personally experienced discrimination can alter the salience of group boundaries and perceptions of belonging without producing a single uniform identity response.

This is particularly relevant to intergroup contact because contact itself can communicate inclusion and belonging. Among ethnic-minority adolescents in the Netherlands, Munniksma et al. (2015) found that friendships with majority-group peers were associated with stronger host-society identification, which was subsequently related to more favorable attitudes toward the majority group. Perceived discrimination can communicate something quite different: that the same ethnic or religious group boundary continues to structure unequal treatment. Thus, frequent cross-group contact and perceived discrimination may coexist while providing conflicting information about the quality and equality of intergroup relations.

Evidence from adolescent school contexts supports this broader contextual interpretation. Tropp et al. (2016) found that perceived discrimination was associated with

lower comfort in cross-ethnic interaction and, among Latino students, lower contact quality. Their longitudinal findings further showed that supportive school norms predicted more positive interethnic experiences under some conditions of lower perceived discrimination, whereas higher discrimination could conflict with institutional messages promoting positive cross-group relations. Although these findings do not directly test whether perceived discrimination moderates the relationship between contact and acceptance of discrimination, they demonstrate that discriminatory experiences can qualify how otherwise favorable intergroup signals are interpreted.

The direct relationship between perceived discrimination and acceptance of discrimination, however, is less straightforward. Experiencing discrimination may heighten awareness of unfair treatment and group-based inequality, but it does not necessarily generalize into greater rejection of discrimination toward other groups. Neumann and Moy (2018), for example, found limited evidence that personally experienced discrimination consistently translated into more inclusive immigration attitudes. Conversely, Steeman et al. (2026) found that personal discrimination experiences were positively associated with acceptance of discrimination in a Belgian adult sample, whereas intergroup contact was negatively associated with acceptance. These contrasting possibilities provide little basis for a directional hypothesis concerning the main association between perceived discrimination and discrimination acceptance, which is therefore examined exploratorily.

The rationale for moderation is more specific. More frequent intergroup contact may be associated with lower acceptance of discrimination because repeated interaction across group boundaries can reduce social distance and provide interpersonal evidence of inclusion. However, as perceived discrimination within that same domain increases, adolescents simultaneously have more personally relevant evidence that the group boundary continues to structure unequal treatment. Under these conditions, frequent intergroup contact may provide less clear evidence that intergroup relations are characterized by equality and inclusion. Accordingly, the negative association between contact and acceptance of discrimination is expected to become weaker at higher levels of perceived discrimination. Because the meaning of discrimination is tied to the group membership to which the unfair treatment is attributed, this moderation is expected to operate within the corresponding ethnic or religious domain.

Taken together, these findings suggest that lived experiences with discrimination, both direct and vicarious, may condition and moderate the relationship between intergroup contact and adolescents' acceptance of discrimination. Exposure to discriminatory events can shape

perceptions of intergroup relations, influence identity processes, and alter expectations regarding fairness and equality. In contexts where discrimination is frequently observed or personally experienced, intergroup contact may be less effective in reducing the perceived legitimacy or acceptability of discriminatory practices.

3.6 The Present Study: Research Question and Hypotheses

The literature reviewed above suggests that intergroup contact is generally associated with more favorable intergroup attitudes, but that this association is shaped by the social conditions in which contact occurs. Contact can reduce social distance, increase familiarity, and foster more inclusive orientations toward outgroups (Pettigrew & Tropp, 2006). Importantly, research also shows that the implications of contact depend on factors such as group status, discrimination, and the normative environment surrounding intergroup interaction (Cocco et al., 2024; Tropp et al., 2016). In the present study, these insights are extended from prejudice and outgroup attitudes to normative judgments about discriminatory treatment. More frequent interaction with members of an ethnic or religious outgroup is therefore expected to be associated with a lower likelihood of accepting discrimination against that group. This expectation is also consistent with Belgian research linking greater domain-relevant intergroup contact to lower acceptance of discrimination (Steeman et al., 2026).

At the same time, the association between contact and acceptance is not expected to be uniform across adolescents. Migrant background, vicarious discrimination, and perceived discrimination represent theoretically distinct conditions under which contact is experienced and interpreted. Migrant background captures differences in social position within intergroup relations. Majority- and minority-background adolescents may encounter cross-group interaction under different conditions of status, belonging, and exposure to discrimination. Research among adolescents shows that school norms, peer norms, and discrimination relate differently to interethnic experiences among ethnic-majority and ethnic-minority youth (Tropp et al., 2016), while contact among minority adolescents may also be connected to identification with the broader host society (Munniksma et al., 2015). Accordingly, the negative association between contact and acceptance is expected to be weaker among adolescents with a migrant background than among those with a majority background.

Vicarious and perceived discrimination may qualify the contact association through somewhat different processes. Vicarious discrimination provides information about intergroup relations beyond adolescents' own interpersonal encounters. Witnessing discriminatory treatment can make broader patterns of inequality visible (Uluğ & Tropp, 2021), while the social responses surrounding discrimination can influence how observers interpret such treatment (Sanchez et al., 2024). Thus, adolescents may experience frequent cross-group contact while simultaneously observing evidence that group membership continues to structure unequal treatment. In such circumstances, frequent intergroup contact may be less indicative of intergroup relations characterized by equality and inclusion. The negative association between contact and acceptance is therefore expected to be weaker among adolescents who report vicarious discrimination than among those who do not.

Perceived discrimination, by contrast, provides personally relevant evidence of unequal treatment within a particular group domain. Perceiving oneself as a target of discrimination can affect how individuals understand their social position, belonging, and group membership (Jasinskaja-Lahti et al., 2009; Ramos et al., 2012; Schmitt et al., 2014). Research with adolescents further shows that higher perceived discrimination is associated with lower comfort in cross-ethnic interaction and that discrimination can alter how supportive intergroup norms relate to contact experiences (Tropp et al., 2016). Consequently, as perceived discrimination within an ethnic or religious domain increases, frequent contact within that same domain may become a less consistent signal of equality and inclusion. The negative association between domain-specific contact and acceptance is therefore expected to become weaker at higher levels of perceived domain-specific discrimination.

These moderation predictions do not imply clear directional expectations for the direct associations of migrant background or discrimination experiences with acceptance. Witnessing discrimination can increase awareness of inequality and motivate responses against injustice, but its consequences are not uniform. Likewise, personally experiencing discrimination may heighten the salience of unfair treatment and group boundaries without necessarily producing greater inclusivity toward other groups. Existing research provides mixed evidence in this respect: discrimination does not consistently generalize into more inclusive attitudes toward other groups, and some research has found personal discrimination experiences to be associated with greater acceptance of discrimination (Neumann & Moy, 2018; Steeman et al., 2026). Migrant background is similarly heterogeneous and encompasses different ethnic, religious, national, generational, and socioeconomic positions. Because these competing processes do not support clear directional predictions, the direct associations of

migrant background, perceived discrimination, and vicarious discrimination with acceptance are examined exploratorily.

Taken together, the present study examines whether domain-specific intergroup contact is associated with adolescents' acceptance of ethnic and religious discrimination and whether this association varies according to migrant background, vicarious discrimination, and perceived domain-specific discrimination. These moderators capture different dimensions of the intergroup context: social position, observed unequal treatment, and personally experienced unequal treatment, respectively. Because the study relies on cross-sectional observational data, the hypotheses concern statistical associations rather than causal effects. The same theoretical expectations are tested separately for ethnic and religious discrimination.

H1: Within each domain, more frequent domain-specific intergroup contact will be associated with a lower likelihood of expressing any acceptance of discrimination.

H2: Within each domain, the negative association between domain-specific intergroup contact and acceptance of discrimination will be weaker among adolescents with a migrant background than among adolescents with a majority background.

H3: Within each domain, the negative association between domain-specific intergroup contact and acceptance of discrimination will be weaker among adolescents who report vicarious discrimination than among adolescents who do not.

H4: Within each domain, the negative association between domain-specific intergroup contact and acceptance of discrimination will become weaker as perceived domain-specific discrimination increases.

Exploratory research question: How are migrant background, perceived domain-specific discrimination, and vicarious discrimination associated with adolescents' likelihood of expressing any acceptance of ethnic and religious discrimination?

4 Methods

4.1 Research design

This study uses a quantitative, cross-sectional research design to examine how intergroup contact and discrimination experiences are associated with adolescents' acceptance of ethnic and religious discrimination. The analysis focuses on adolescents in Dutch-speaking Belgian schools and uses survey data containing measures of contact with outgroups, perceived unequal treatment, vicarious discrimination, migrant background, and attitudes toward discrimination.

The empirical strategy is based on two parallel domain-specific analyses. Rather than treating ethnic and religious discrimination as one combined outcome, the study analyzes them separately. This decision follows the conceptual distinction between ethnic and religious discrimination and the empirical concern that the items are related but not interchangeable. The two-domain approach makes it possible to examine whether the relationship between contact, discrimination experiences, and acceptance of discrimination works similarly or differently across ethnic and religious domains.

The analysis is observational and cross-sectional. Therefore, the models do not establish causal effects. Instead, they estimate associations between intergroup contact, discrimination experiences, migrant background, and the likelihood that adolescents express any acceptance of discrimination. The wording throughout the results and discussion therefore refers to associations rather than causal impacts.

4.2 Data and sample

The analysis is based on the same dataset used in De Coninck, Steeman, and d'Haenens' study on Belgian adolescents' acceptance of discrimination in the digital age (2025a). The dataset was provided by Dr. De Coninck for the purpose of conducting a secondary analysis.

The original data were collected through an online survey administered to pupils in the third cycle of Dutch-language secondary education in Belgium, corresponding mainly to

adolescents aged 16 to 18. The survey was programmed in Qualtrics and administered during school visits between February and April 2025.

The original study used a two-stage sampling procedure. In the first stage, secondary schools were selected through quota sampling, with quotas based on educational orientation and province. In the second stage, participating schools selected classes to participate in the survey. The aim was to include two classes from both the fifth and sixth years of secondary education across each educational orientation: transfer orientation, double orientation, and labor market orientation.

In total, 815 pupils participated in the original data collection. Following the original study's data-quality procedure, 53 respondents who completed less than 80% of the survey were excluded. The final dataset therefore consisted of 762 respondents.

Because this thesis uses the dataset for a different analytical purpose than the original paper, the variables and models were re-specified. Whereas the original paper used latent profile analysis to identify profiles of discrimination acceptance across multiple grounds, this thesis focuses specifically on ethnic and religious discrimination as two separate domains. The analysis therefore estimates domain-specific logistic regression models rather than latent profile models.

4.3 Analytical sample and missing data

The full dataset contained 762 respondents. The regression analyses required valid information on the dependent variables, domain-specific intergroup contact, perceived discrimination, migrant background, vicarious discrimination, gender, age, mother's education, and school location. Complete-case analysis was therefore used for the primary models. Respondents were retained only when information was available for every variable included in the final model sequence.

Complete information on the core ethnic-domain variables was available for 617 respondents, and the same number had complete information on the corresponding religious-domain variables. After the control variables were included in the complete-case requirement, the final analytic sample consisted of 517 respondents in both domains. All primary models within each domain were estimated using this common sample. Holding the sample constant

ensured that Akaike information criterion values and likelihood-ratio tests reflected differences in model specification rather than changes in the respondents included.

Missingness was highest for mother's education, for which 146 cases, or 19.2% of the full dataset, were missing. Migrant background was missing for 93 respondents, and vicarious discrimination was missing for 74 respondents. Missingness was considerably lower for the acceptance outcomes and contact measures. Each acceptance outcome had 13 missing observations, while each domain-specific contact measure had seven.

Because mother's education accounted for substantial case loss, sensitivity analyses were estimated without this control. Excluding mother's education from the complete-case requirement increased both domain-specific samples from 517 to 601 respondents, recovering 84 cases, or 16.2%. Mother's education was nevertheless retained in the primary models because it was treated as a theoretically relevant indicator of socioeconomic background. The larger samples were used to assess whether the main conclusions depended on this control and the associated reduction in sample size.

Complete-case analysis may introduce selection bias when respondents with missing information differ systematically from those retained. The sensitivity analyses address the principal source of missingness but do not eliminate this possibility. The results are therefore interpreted with appropriate caution.

4.4 Analytical focus: ethnic and religious domains

The analysis distinguishes between ethnic and religious discrimination as related but analytically separate domains. Ethnic discrimination concerns unequal treatment based on characteristics such as ethnicity, race, nationality, origin, or skin colour, whereas religious discrimination concerns unequal treatment based on religious denomination or identity. Although these forms of discrimination may overlap socially, particularly for some migrant-background adolescents, they are not assumed to operate identically.

Two parallel sets of models were therefore estimated. In the ethnic domain, acceptance of ethnic discrimination was examined in relation to ethnic intergroup contact and perceived ethnic discrimination. In the religious domain, acceptance of religious discrimination was examined in relation to religious intergroup contact and perceived religious discrimination. Migrant background, vicarious discrimination, and the control

variables were included in both sets of models. This domain-matching strategy ensures that each acceptance outcome is examined using the corresponding contact and perceived-discrimination measures.

4.5 Measures

4.5.1 *Dependent variables*

The dependent variables measure adolescents' acceptance of ethnic and religious discrimination. Acceptance of ethnic discrimination was measured using Q20_1, which concerns discrimination based on ethnicity, race, nationality, or skin colour. Acceptance of religious discrimination was measured using Q20_4, which concerns discrimination based on religious denomination. Both items were originally measured on a seven-point scale ranging from 1, indicating that discrimination is never acceptable, to 7, indicating that discrimination is always acceptable.

Responses to both items were strongly concentrated in the lowest category, producing pronounced floor effects. For the primary analyses, each outcome was therefore recoded as a binary variable distinguishing complete rejection from any degree of acceptance. Respondents who selected 1 were coded as 0, indicating complete rejection of discrimination, while respondents who selected a value from 2 to 7 were coded as 1, indicating some level of acceptance.

Because binary coding reduces variation within the original response scale, supplementary ordinal analyses were conducted using the original seven response categories. These sensitivity analyses used cumulative-link models to assess whether the principal findings depended on distinguishing complete rejection from any acceptance. For the ethnic outcome, the proportional-odds assumption was relaxed for gender after the diagnostic test indicated that this predictor did not satisfy the assumption. The religious outcome was estimated using the standard proportional-odds specification.

4.5.2 *Intergroup contact*

The main independent variable is domain-specific intergroup contact. Ethnic intergroup contact was measured using Q15_1, which asked respondents how often they had interacted during the previous month with people of another ethnicity, race, nationality, or skin colour. Religious intergroup contact was measured using Q15_7, which asked about interaction with people of another religious denomination. Both items were measured on a six-point scale, with higher scores indicating more frequent intergroup contact.

The contact variables were treated as continuous predictors. Before model estimation, each variable was mean-centred within its corresponding analytic sample. Mean-centring does not alter the estimated association between contact and the outcome, but it facilitates the interpretation of coefficients in models containing interaction terms. When contact was interacted with a continuous moderator, the moderator was also mean-centred. The main effect of contact could therefore be interpreted at the mean value of that moderator. When contact was interacted with a categorical moderator, its main effect represented the association for the moderator's reference category.

4.5.3 *Perceived discrimination*

Perceived discrimination was measured separately for the ethnic and religious domains. Perceived ethnic discrimination was measured using Q16_1, which captures respondents' perceptions of unequal treatment based on ethnicity, nationality, race, or skin colour. Perceived religious discrimination was measured using Q16_4, which captures perceived unequal treatment based on religion. Higher scores indicated greater perceived discrimination.

Both variables were treated as continuous predictors. They were included as main effects to assess whether perceived discrimination was associated with acceptance of discrimination within the corresponding domain. They were also examined as moderators of the association between domain-specific intergroup contact and acceptance of discrimination.

Before estimating the interaction models, each perceived-discrimination variable was mean-centred within its corresponding analytic sample. In models containing the interaction between contact and perceived discrimination, centring allowed the main effect of contact to be interpreted at the sample mean of perceived discrimination.

4.5.4 *Vicarious discrimination*

Vicarious discrimination was measured using Q18 and Q19, which assessed whether respondents had witnessed or encountered discrimination directed at other people in person or online. The two items were combined into a binary indicator. Respondents who answered yes to either item were coded as 1, indicating exposure to vicarious discrimination. Respondents who answered no to both items were coded as 0. Cases with insufficient information to determine exposure were coded as missing.

Vicarious discrimination was included as a main predictor in both the ethnic and religious models. It was also examined as a moderator of the association between domain-specific intergroup contact and acceptance of discrimination. In the interaction models, respondents who did not report vicarious discrimination formed the reference category.

4.5.5 *Migrant background*

Migrant background was constructed using respondents' country of birth and parental country of birth. Q42 measured where the respondent was born, while Q43 measured the birth countries of the respondent's parents. Because the imported dataset codes did not directly correspond to the original questionnaire categories, both variables were first recoded to their original response categories.

Respondents were coded as having a majority background when they were born in Belgium and both parents were also born in Belgium. Respondents were coded as having a migrant background when they were born outside Belgium or when at least one parent was born outside Belgium. Cases with insufficient information to determine background were coded as missing.

Migrant background was included as a main predictor in both domain-specific models and as a moderator of the association between intergroup contact and acceptance of discrimination. Majority-background adolescents formed the reference category. This binary indicator captures migration background broadly and does not distinguish between particular countries of origin, migrant generations, ethnic identities, or religious affiliations.

4.5.6 Control variables

The adjusted models included gender, age, mother's education, and school location as control variables. Gender, age, and mother's education were treated as individual-level demographic controls, while school location was included as a geographical control. These variables were selected because they may be associated with adolescents' opportunities for intergroup contact, experiences of discrimination, and normative evaluations of unequal treatment.

Gender was measured using Q32 and coded categorically, with men serving as the reference category. Age was measured using Q33. Responses corresponding to ages 15 through 19 were converted to their numerical values and treated as a continuous variable. Responses in the residual "other" category were coded as missing because they did not provide a specific age.

Mother's education was measured using Q37 and grouped into three substantive categories: low education, comprising no education or primary education; secondary education; and higher education, comprising college, university, or doctoral education. Secondary education served as the reference category. Mother's education was included as an approximate indicator of socioeconomic background.

School location was measured using Q38 and classified as city, suburb, village, or countryside. City schools formed the reference category. This variable captures the broad geographical setting of the school rather than characteristics of individual schools or classrooms.

The control variables were included to reduce potential confounding in the estimated associations. Their coefficients were not treated as the primary substantive focus of the study.

4.6 Analytical strategy

All analyses were conducted in R. Descriptive analyses were first used to examine the distributions of the study variables. Frequencies and percentages were calculated for categorical variables, while means, standard deviations, medians, and ranges were calculated for continuous and ordinal variables. Spearman correlations were used to describe the bivariate relationships among the domain-specific contact, perceived discrimination, and acceptance measures.

The primary analyses used binary logistic regression because the dependent variables distinguished respondents who completely rejected discrimination from those who expressed any level of acceptance. Separate model sequences were estimated for the ethnic and religious domains. The models estimated the association between each predictor and the odds of expressing any acceptance of discrimination.

The analysis proceeded in stages. First, a contact-only model examined the unadjusted association between domain-specific intergroup contact and acceptance of discrimination. Second, an uncontrolled main-effects model added perceived discrimination, migrant background, and vicarious discrimination. Third, the primary controlled main-effects model additionally adjusted for gender, age, mother's education, and school location.

Moderation was assessed by adding interaction terms between contact and each proposed moderator: perceived discrimination, migrant background, and vicarious discrimination. Each interaction was first estimated in a separate controlled model to preserve interpretability. A model containing all three interactions was also estimated as a robustness check. Interaction models were compared with the corresponding controlled main-effects model using likelihood-ratio tests and the Akaike information criterion. Lower AIC values indicated better relative model fit, while a statistically significant likelihood-ratio test indicated that the interaction improved fit relative to the simpler model.

Logistic regression coefficients were exponentiated and presented as odds ratios with 95% confidence intervals. An odds ratio below 1 indicated that higher values of the predictor were associated with lower odds of expressing any acceptance of discrimination, while an odds ratio above 1 indicated higher odds. Statistically significant interactions were examined using simple slopes and predicted probabilities to clarify how the association between contact and acceptance differed across levels or categories of the moderator. Non-significant interaction terms were interpreted as insufficient evidence that the association differed according to the proposed moderator.

The assumptions and stability of the primary controlled models were assessed using several diagnostics. Multicollinearity was evaluated using generalized variance inflation factors. The linearity of continuous predictors in the logit was assessed by comparing the primary models with models containing quadratic terms. Potentially influential observations were examined using Cook's distance, leverage values, and residuals.

Two sets of sensitivity analyses were conducted. First, the primary controlled models and the ethnic contact-by-vicarious-discrimination interaction were re-estimated in larger samples of 601 respondents that did not require complete information on mother's education.

Second, cumulative-link models were estimated using the original seven-category acceptance outcomes to assess whether the conclusions depended on binary recoding. The proportional-odds assumption was assessed for the ordinal models. For the ethnic outcome, gender was permitted to have non-proportional effects because it did not satisfy this assumption, while the religious model retained the standard proportional-odds specification.

All primary models were estimated using the same complete-case sample of 517 respondents within each domain, ensuring that comparisons between models were not affected by changes in sample composition. Because the data are cross-sectional, the estimated relationships are interpreted as associations rather than causal effects.

5 Results

5.1 Sample characteristics and descriptive findings

The primary analytic sample consisted of 517 respondents in both the ethnic and religious domains. Because the same complete-case requirements were applied across the two domains, the demographic composition of both analytic samples was identical. Table 1 summarizes the characteristics of the primary analytic sample and the descriptive statistics for the principal study variables. Respondents had a mean age of 17.00 years ($SD = 0.80$), with ages ranging from 15 to 19 years. Women comprised 62.9% of the sample ($n = 325$), while men comprised 37.1% ($n = 192$).

Most respondents had a majority background ($n = 340$, 65.8%), while 177 respondents (34.2%) had a migrant background. Approximately two-thirds of respondents reported exposure to vicarious discrimination ($n = 342$, 66.2%), while 175 respondents (33.8%) did not. Regarding mother's education, 360 respondents (69.6%) were classified in the higher-education category, 106 (20.5%) in the secondary-education category, and 51 (9.9%) in the low-education category. Nearly half of the respondents attended schools located in villages ($n = 243$, 47.0%), followed by cities ($n = 205$, 39.7%), suburbs ($n = 47$, 9.1%), and the countryside ($n = 22$, 4.3%).

Complete rejection was the most common response in both discrimination domains. For ethnic discrimination, 341 respondents (66.0%) completely rejected discrimination, while 176 (34.0%) expressed some level of acceptance. For religious discrimination, 327 respondents (63.2%) completely rejected discrimination, while 190 (36.8%) expressed some

level of acceptance. The original seven-category outcomes were strongly concentrated at the lowest response value. Mean acceptance was 1.83 (SD = 1.51) for ethnic discrimination and 1.87 (SD = 1.50) for religious discrimination, with a median of 1 in both domains.

Respondents generally reported relatively frequent intergroup contact. Mean ethnic intergroup contact was 4.76 (SD = 1.59) on the six-point scale, while mean religious intergroup contact was 4.70 (SD = 1.65). Perceived discrimination scores were comparatively low, with a mean of 1.60 (SD = 1.01) for perceived ethnic discrimination and 1.46 (SD = 0.96) for perceived religious discrimination.

Table 1: Characteristics of the primary analytic sample

Characteristic	Category or statistic	Value
Age	Mean (SD), range	17.00 (0.80), 15-19
Gender	Man	192 (37.1%)
Gender	Woman	325 (62.9%)
Migrant background	Majority background	340 (65.8%)
Migrant background	Migrant background	177 (34.2%)
Vicarious discrimination	No vicarious discrimination	175 (33.8%)
Vicarious discrimination	Vicarious discrimination	342 (66.2%)
Mother's education	Secondary education	106 (20.5%)
Mother's education	Low education	51 (9.9%)
Mother's education	Higher education	360 (69.6%)
School environment	City	205 (39.7%)
School environment	Suburb	47 (9.1%)
School environment	Village	243 (47.0%)
School environment	Countryside	22 (4.3%)
Ethnic intergroup contact	Mean (SD)	4.76 (1.59)
Religious intergroup contact	Mean (SD)	4.70 (1.65)
Perceived ethnic discrimination	Mean (SD)	1.60 (1.01)
Perceived religious discrimination	Mean (SD)	1.46 (0.96)
Ethnic-discrimination acceptance score	Mean (SD), median	1.83 (1.51), median = 1
Religious-discrimination acceptance score	Mean (SD), median	1.87 (1.50), median = 1
Ethnic-discrimination outcome	Complete rejection	341 (66.0%)
Ethnic-discrimination outcome	Any acceptance	176 (34.0%)
Religious-discrimination outcome	Complete rejection	327 (63.2%)
Religious-discrimination outcome	Any acceptance	190 (36.8%)

Note: N = 517. Values are n (%) unless otherwise indicated. Intergroup contact was measured from 1 to 6, perceived discrimination from 1 to 5, and the original acceptance outcomes from 1 to 7.

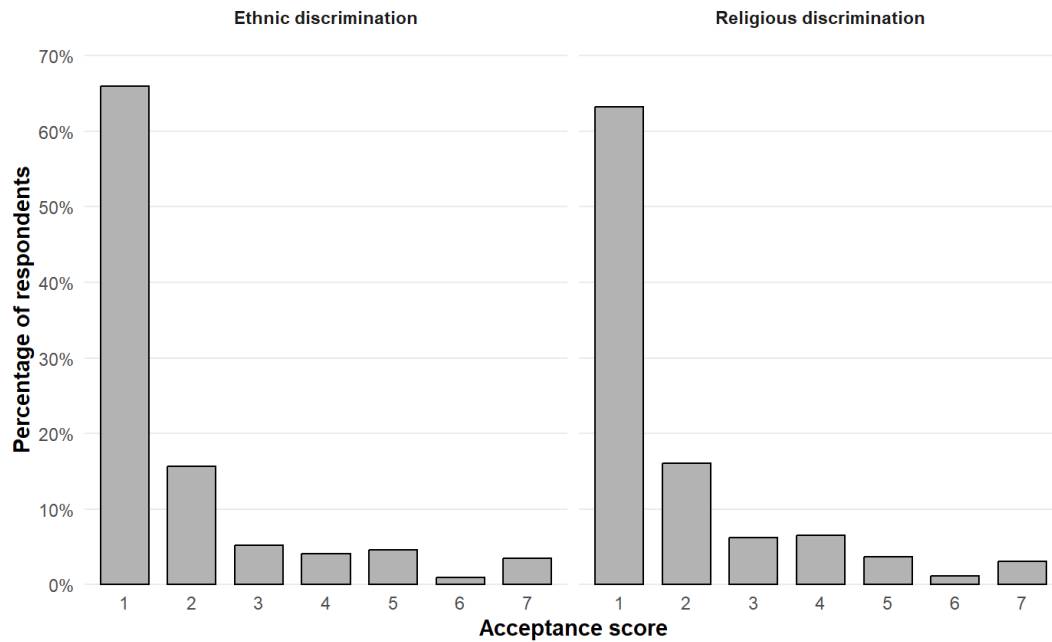


Figure 1. Distribution of the original ethnic- and religious-discrimination acceptance outcomes in the primary analytic sample.

Note: Scores range from 1, never acceptable, to 7, always acceptable. $N = 517$ in each domain.

5.2 Bivariate associations

Spearman correlations indicated modest negative associations between intergroup contact and acceptance of discrimination in both domains. Ethnic intergroup contact was negatively correlated with acceptance of ethnic discrimination, Spearman's $\rho = -.108$, $p = .014$. Religious intergroup contact showed a somewhat stronger negative correlation with acceptance of religious discrimination, Spearman's $\rho = -.155$, $p < .001$.

Perceived ethnic discrimination was not meaningfully correlated with acceptance of ethnic discrimination, Spearman's $\rho = .004$. By contrast, perceived religious discrimination was negatively correlated with acceptance of religious discrimination, Spearman's $\rho = -.147$. Contact and perceived discrimination were positively correlated within both domains, although these associations were modest. The correlation was Spearman's $\rho = .245$ in the ethnic domain and Spearman's $\rho = .196$ in the religious domain.

These bivariate associations provide an initial descriptive overview. The multivariable regression models reported below assess whether the relationships remain after accounting for the other predictors and control variables.

5.3 Primary main-effects models

Table 2 presents the primary controlled logistic regression models for acceptance of ethnic and religious discrimination. Both models adjusted for gender, age, mother's education, and school location.

In the ethnic domain, more frequent ethnic intergroup contact was associated with lower odds of expressing any acceptance of ethnic discrimination, OR = 0.83, 95% CI [0.73, 0.94], $p = .003$. Each one-unit increase in contact corresponded to approximately 17% lower odds of expressing any acceptance. Vicarious discrimination was positively associated with the outcome, OR = 1.98, 95% CI [1.29, 3.07], $p = .002$. Respondents who reported vicarious discrimination had approximately twice the odds of expressing some acceptance compared with those who did not report it. Perceived ethnic discrimination, OR = 0.96, 95% CI [0.75, 1.21], $p = .710$, and migrant background, OR = 1.06, 95% CI [0.65, 1.71], $p = .823$, were not significantly associated with acceptance of ethnic discrimination.

A similar pattern was observed in the religious domain. More frequent religious intergroup contact was associated with lower odds of expressing any acceptance of religious discrimination, OR = 0.83, 95% CI [0.74, 0.93], $p = .002$. This also corresponded to approximately 17% lower odds for each one-unit increase in contact. Vicarious discrimination was associated with higher odds of expressing some acceptance, OR = 1.83, 95% CI [1.21, 2.80], $p = .005$. Perceived religious discrimination was associated with lower odds of acceptance, OR = 0.75, 95% CI [0.58, 0.95], $p = .023$. Migrant background was not significantly associated with acceptance of religious discrimination, OR = 0.87, 95% CI [0.55, 1.37], $p = .561$.

Overall, the results supported the hypothesis that more frequent domain-specific intergroup contact would be associated with lower acceptance of discrimination in both domains. The findings also indicated a consistent positive association between vicarious discrimination and acceptance, while the association involving perceived discrimination differed between the ethnic and religious domains.

Table 2: Adjusted logistic regression models predicting any acceptance of discrimination

Predictor	Ethnic OR [95% CI]	Ethnic p	Religious OR [95% CI]	Religious p
Intergroup contact	0.83 [0.73, 0.94]	.003	0.83 [0.74, 0.93]	.002
Perceived discrimination	0.96 [0.75, 1.21]	.710	0.75 [0.58, 0.95]	.023
Migrant background	1.06 [0.65, 1.71]	.823	0.87 [0.55, 1.37]	.561
Vicarious discrimination	1.98 [1.29, 3.07]	.002	1.83 [1.21, 2.80]	.005
Woman	0.43 [0.29, 0.64]	< .001	0.58 [0.40, 0.86]	.007
Age	1.00 [0.78, 1.28]	.974	0.98 [0.77, 1.25]	.887
Mother's education: low	1.95 [0.96, 3.97]	.065	1.21 [0.58, 2.50]	.607
Mother's education: higher	0.69 [0.41, 1.14]	.146	0.70 [0.42, 1.15]	.159
School environment: suburb	1.14 [0.57, 2.27]	.702	1.56 [0.80, 3.07]	.191
School environment: village	1.11 [0.73, 1.70]	.626	1.14 [0.76, 1.73]	.530
School environment: countryside	0.70 [0.23, 1.91]	.508	0.72 [0.24, 1.90]	.523

Note: Cells report odds ratios with 95% confidence intervals. The outcomes are coded 1 for any acceptance and 0 for complete rejection. Both models use $N = 517$. Reference categories are majority background, no vicarious discrimination, man, secondary maternal education, and city school location.

5.4 Moderation analyses

Table 3 presents the likelihood-ratio tests comparing each controlled interaction model with the corresponding controlled main-effects model. There was no evidence that the association between intergroup contact and acceptance differed according to migrant background. Adding the contact-by-migrant-background interaction did not improve model fit in either the ethnic domain, $\chi^2(1) = 0.02$, $p = .901$, or the religious domain, $\chi^2(1) = 0.09$, $p = .764$. Hypothesis 2 was therefore not supported.

Perceived discrimination also did not significantly moderate the association between contact and acceptance. The contact-by-perceived-discrimination interaction did not improve model fit for ethnic discrimination, $\chi^2(1) = 0.15$, $p = .702$, or religious discrimination, $\chi^2(1) = 1.70$, $p = .192$. Hypothesis 4 was therefore not supported.

Evidence of moderation by vicarious discrimination was found only in the ethnic domain. The ethnic contact-by-vicarious-discrimination interaction was statistically significant, $OR = 1.30$, 95% CI [1.01, 1.68], $p = .040$, and improved model fit relative to the controlled main-effects model, $\chi^2(1) = 4.23$, $p = .040$. Because the interaction odds ratio was

above 1, the negative association between ethnic contact and acceptance became weaker among respondents who reported vicarious discrimination.

Simple-slope analyses clarified this pattern. Among respondents who did not report vicarious discrimination, greater ethnic contact was associated with lower odds of accepting ethnic discrimination, $OR = 0.69$, 95% CI $[0.56, 0.86]$, $p < .001$. Among respondents who reported vicarious discrimination, the association was weaker and not statistically significant, $OR = 0.90$, 95% CI $[0.78, 1.04]$, $p = .164$. The significant interaction indicates that the two slopes differed, rather than merely reflecting that one slope was statistically significant and the other was not. Figure 2 illustrates the adjusted predicted probabilities underlying this interaction.

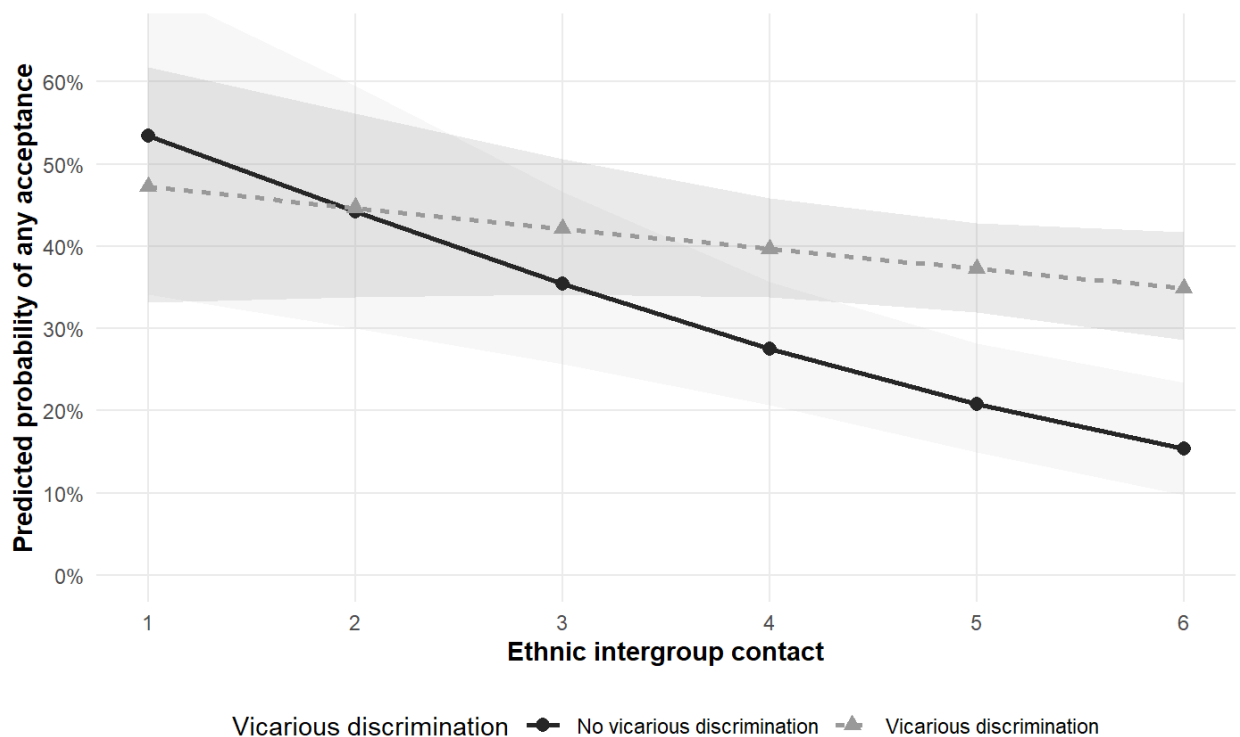


Figure 2. Adjusted predicted probability of any acceptance of ethnic discrimination by ethnic intergroup contact and vicarious-discrimination status.

Note: Predictions are derived from the primary controlled interaction model, $N = 517$. Shaded areas represent 95% confidence intervals.

Table 3: Likelihood-ratio tests comparing the controlled interaction models with the controlled main-effects models

Domain	Added interaction	LR chi-square	df	p-value
Ethnic	Contact x perceived discrimination	0.15	1	.702
Ethnic	Contact x migrant background	0.02	1	.901
Ethnic	Contact x vicarious discrimination	4.23	1	.040
Ethnic	All three interactions	4.41	3	.220
Religious	Contact x perceived discrimination	1.70	1	.192
Religious	Contact x migrant background	0.09	1	.764
Religious	Contact x vicarious discrimination	1.22	1	.269
Religious	All three interactions	3.11	3	.376

Note: Each interaction model was compared with the corresponding controlled main-effects model. All models use $N = 517$.

The corresponding interaction was not significant in the religious domain, $\chi^2(1) = 1.22$, $p = .269$. The models containing all three interactions simultaneously also did not improve fit in either the ethnic domain, $\chi^2(3) = 4.41$, $p = .220$, or the religious domain, $\chi^2(3) = 3.11$, $p = .376$. Hypothesis 3 therefore received partial support, limited to acceptance of ethnic discrimination.

5.5 Model diagnostics

Diagnostic checks indicated that the primary logistic regression models were not affected by serious multicollinearity. Adjusted generalized variance inflation factors ranged from 1.01 to 1.23, remaining well below conventional thresholds for concern.

There was also no evidence that the continuous predictors violated the assumption of linearity in the logit. Adding quadratic terms did not significantly improve model fit in either the ethnic domain, $\chi^2(3) = 4.40$, $p = .221$, or the religious domain, $\chi^2(3) = 3.91$, $p = .272$.

Influence diagnostics identified no observations with a Cook's distance above 1 and no standardized residuals with an absolute value above 3. The maximum Cook's distances were .019 for the primary ethnic model, .021 for the primary religious model, and .021 for the ethnic contact-by-vicarious-discrimination model. Although some observations exceeded conventional screening thresholds for leverage or Cook's distance, none appeared sufficiently influential to justify exclusion. All observations were therefore retained.

Taken together, these diagnostics provided no indication that multicollinearity, non-linearity, or influential cases materially affected the primary findings.

5.6 Sensitivity analyses

The first sensitivity analysis examined whether the findings depended on including mother's education and the associated complete-case restriction. Excluding mother's education increased the analytic sample from 517 to 601 respondents in both domains.

The principal main-effect findings remained substantively similar in the larger samples. Ethnic intergroup contact remained associated with lower odds of accepting ethnic discrimination, $OR = 0.86$, 95% CI $[0.77, 0.96]$, $p = .008$, while vicarious discrimination remained associated with higher odds, $OR = 1.76$, 95% CI $[1.20, 2.60]$, $p = .004$. Perceived ethnic discrimination and migrant background remained non-significant.

In the religious domain, intergroup contact remained negatively associated with acceptance, $OR = 0.87$, 95% CI $[0.78, 0.97]$, $p = .011$. Perceived religious discrimination also remained negatively associated with acceptance, $OR = 0.78$, 95% CI $[0.62, 0.98]$, $p = .034$, while vicarious discrimination remained positively associated, $OR = 1.77$, 95% CI $[1.22, 2.60]$, $p = .003$. Migrant background remained non-significant.

The ethnic contact-by-vicarious-discrimination interaction also remained statistically significant in the larger sample, $OR = 1.26$, 95% CI $[1.00, 1.59]$, $p = .047$. Among respondents who did not report vicarious discrimination, ethnic contact was associated with lower odds of acceptance, $OR = 0.74$, 95% CI $[0.61, 0.89]$, $p = .001$. Among those who reported vicarious discrimination, the association was weaker and non-significant, $OR = 0.93$, 95% CI $[0.81, 1.07]$, $p = .305$. This pattern was consistent with the primary analysis.

A second sensitivity analysis used the original seven-category acceptance outcomes in cumulative-link models. The ordinal results closely resembled the binary logistic regression findings. In the ethnic domain, greater contact was associated with lower acceptance, $OR = 0.82$, 95% CI $[0.73, 0.93]$, $p = .001$, while vicarious discrimination was associated with higher acceptance, $OR = 1.95$, 95% CI $[1.28, 2.98]$, $p = .002$. Perceived ethnic discrimination and migrant background were not significant.

In the religious ordinal model, greater contact was again associated with lower acceptance, $OR = 0.85$, $p = .002$. Perceived religious discrimination was negatively associated with acceptance, $OR = 0.74$, $p = .016$, while vicarious discrimination was positively associated, $OR = 1.82$, $p = .004$. Migrant background remained non-significant.

Overall, the sensitivity analyses indicated that the main conclusions were not dependent on including mother's education in the model or on dichotomizing the acceptance

outcomes. Table 4 compares the focal estimates across the primary binary models and the two sensitivity specifications.

Table 4: Comparison of primary and sensitivity estimates for the focal predictors

Domain	Predictor	Primary binary (N = 517)	Excludes mother's education (N = 601)	Ordinal outcome (N = 517)
Ethnic	Intergroup contact	0.83 [0.73, 0.94]	0.86 [0.77, 0.96]	0.82 [0.73, 0.93]
Ethnic	Perceived discrimination	0.96 [0.75, 1.21]	0.93 [0.75, 1.15]	0.99 [0.79, 1.24]
Ethnic	Migrant background	1.06 [0.65, 1.71]	1.16 [0.76, 1.79]	1.04 [0.65, 1.65]
Ethnic	Vicarious discrimination	1.98 [1.29, 3.07]	1.76 [1.20, 2.60]	1.95 [1.28, 2.98]
Religious	Intergroup contact	0.83 [0.74, 0.93]	0.87 [0.78, 0.97]	0.85 [0.76, 0.94]
Religious	Perceived discrimination	0.75 [0.58, 0.95]	0.78 [0.62, 0.98]	0.74 [0.58, 0.94]
Religious	Migrant background	0.87 [0.55, 1.37]	0.87 [0.58, 1.31]	0.90 [0.58, 1.38]
Religious	Vicarious discrimination	1.83 [1.21, 2.80]	1.77 [1.22, 2.60]	1.82 [1.21, 2.74]

Note: Cells report odds ratios with 95% confidence intervals. The larger binary models exclude mother's education. The ethnic ordinal estimates are derived from the partial proportional-odds model in which gender was permitted to vary across thresholds.

6 Discussion

The present study examined how intergroup contact, perceived discrimination, vicarious discrimination, and migrant background were associated with adolescents' acceptance of ethnic and religious discrimination in Dutch-language Belgian secondary schools. By distinguishing complete rejection of discrimination from any degree of acceptance, the analysis focused on whether adolescents were willing to regard unequal ethnic or religious treatment as acceptable at all. The results produced four principal findings. First, more frequent domain-specific intergroup contact was consistently associated with lower acceptance of both ethnic and religious discrimination, supporting Hypothesis 1. Second, vicarious discrimination was associated with higher acceptance in both domains. Third, perceived religious discrimination was associated with lower acceptance of religious discrimination, whereas perceived ethnic discrimination was not significantly associated with acceptance of ethnic discrimination. Finally, migrant background was related neither to acceptance nor to the association between contact and acceptance. Evidence for moderation was limited to a tentative interaction between ethnic contact and vicarious discrimination. Hypotheses 2 and 4 were therefore not supported, while Hypothesis 3 received partial support only in the ethnic domain. These patterns remained largely stable across the larger-sample and ordinal-outcome sensitivity analyses.

6.1 Intergroup contact and the rejection of discrimination

The most consistent finding was the negative association between intergroup contact and discrimination acceptance. In the controlled models, every one-unit increase in contact was associated with approximately 17% lower odds of expressing any acceptance of discrimination in both the ethnic and religious domains. The similarity of the estimates across the two domains is notable because ethnic and religious contact were measured separately and were matched to distinct outcomes. The result therefore suggests that the association was not limited to one specific type of group boundary.

This finding is consistent with intergroup contact theory, which proposes that interaction with members of other social groups can improve intergroup attitudes. Pettigrew and Tropp's (2006) meta-analysis found that greater contact was generally associated with lower prejudice across a wide variety of populations, group distinctions, and settings. Contact effects also frequently generalized beyond the specific individual encountered to broader evaluations of the outgroup. Conditions such as cooperation, equal status, shared goals, and institutional support strengthened these associations, although they were not necessary for favourable contact outcomes to emerge (Pettigrew & Tropp, 2006)

The present findings extend this literature in an important respect. Most contact research examines prejudice, social distance, outgroup attitudes, friendship interest, or intergroup anxiety. The dependent variable in this study instead concerned a normative evaluation of unequal treatment. Adolescents with more frequent contact were not merely more positive toward an outgroup; they were also less likely to regard discrimination against that group as acceptable. Contact may therefore be related to moral and normative boundaries surrounding unequal treatment, in addition to affective evaluations of outgroups.

Several mechanisms could account for this association. Contact may provide information that challenges simplified stereotypes and increases familiarity with the diversity present within an outgroup. It may also reduce anxiety and perceived social distance or encourage adolescents to view outgroup members as individuals rather than as interchangeable representatives of a social category. Among ethnic-minority adolescents in the Netherlands, friendships with majority-group peers were associated with stronger host-society identification, which was subsequently related to more favourable attitudes toward the majority group (Munniksma et al., 2015). Such processes may make discriminatory treatment harder to justify because the people affected are understood as familiar peers, friends, or members of a shared social environment.

These mechanisms were not directly measured in the present study, however. The contact items captured how frequently adolescents interacted with people from different ethnic or religious groups. They did not assess whether the interaction was positive, voluntary, cooperative, equal in status, or sufficiently close to constitute friendship. It is therefore more accurate to conclude that contact frequency was associated with lower acceptance than to claim that a particular type of positive contact produced the result.

The cross-sectional design also prevents a causal interpretation. Adolescents who categorically reject discrimination may be more willing to enter diverse settings or maintain cross-group relationships. Stable features of schools, families, peer networks, or personality could also contribute both to contact and to egalitarian attitudes. This caution is supported by recent longitudinal research. McKeown et al. (2025) found strong between-person associations between youth contact and several social-cohesion outcomes in Belfast and Bradford, but limited and inconsistent evidence that within-person increases in contact produced later improvement. Similarly, Hooghe et al. (2013) found that Belgian adolescents with intergroup friends had lower initial ethnocentrism, but friendship did not consistently predict later changes in ethnocentrism.

The present result should consequently be interpreted as robust evidence of an association rather than proof of a contact effect. Nevertheless, its replication across ethnic and religious domains and across binary and ordinal specifications strengthens the conclusion that contact is meaningfully connected to adolescents' normative rejection of discrimination.

Vicarious discrimination and greater acceptance

The positive association between vicarious discrimination and acceptance was less expected. Adolescents who reported witnessing or encountering discrimination directed at others had approximately twice the odds of expressing some acceptance of ethnic discrimination and 1.83 times the odds of expressing some acceptance of religious discrimination. These associations persisted in both sensitivity analyses.

Much of the existing literature presents vicarious discrimination as a harmful stressor. Witnessing or learning about discrimination against relatives, friends, or other group members has been associated with depression, anxiety, anger, loneliness, sleep difficulties, and other negative outcomes (Dhanani et al., 2022; Li Verdugo et al., 2024; Louie & Upenieks, 2022). This literature demonstrates that indirect exposure can be psychologically consequential, but it provides little evidence about whether such exposure increases or decreases acceptance of discriminatory behaviour. The present study therefore addresses a different and comparatively underexplored outcome.

One possible interpretation is normalization through repeated exposure. Witnessing discriminatory remarks, exclusion, or hostile treatment may communicate information not only about the target group but also about the norms of the surrounding environment. If discrimination appears frequent and receives little sanction from peers or authorities, adolescents may infer that such behaviour is ordinary, tolerated, or less morally serious than formal equality norms suggest. Adolescents are particularly attentive to peer approval and group norms because social belonging and avoidance of exclusion are highly salient during this developmental period (Tomova et al., 2021). Under these conditions, observing discrimination may contribute to descriptive norms about what people actually do, even when respondents are aware of prescriptive norms stating that discrimination is wrong.

A second possibility is that vicarious exposure is associated with perceived intergroup threat or defensive boundary-drawing rather than solidarity. Witnessing conflict between groups may reinforce the salience of ethnic or religious boundaries and encourage adolescents to view exclusion as understandable or protective. Prior Belgian research shows that experiencing unfair treatment does not invariably translate into broader egalitarianism. In the adolescent profile study by De Coninck et al. (2025a), discrimination experiences were associated with some non-egalitarian profiles. Steeman et al. (2026) similarly found that adults who reported more discrimination also reported greater acceptance of several forms of discrimination. These findings suggest that exposure to unfair treatment may coexist with exclusionary orientations rather than automatically producing solidarity with other marginalized groups.

A third explanation concerns selection and measurement. Adolescents with more accepting attitudes may participate in peer groups, online communities, or social environments in which discrimination is more visible or frequent. They may also be more willing to classify what they see as an ordinary interaction rather than intervene or disengage. Conversely, highly egalitarian adolescents might avoid hostile online spaces or peer environments. The direction of the association cannot be determined from the present data.

The vicarious-discrimination variable also combined in-person and online exposure into a single binary indicator. It did not measure the frequency or severity of exposure, the relationship between the observer and the target, the social group targeted, the observer's emotional reaction, or whether the discrimination was condemned by others. A student who once encountered a discriminatory video online received the same score as one who repeatedly witnessed a close friend being harassed at school. The measure therefore captures broad exposure but cannot identify the process connecting exposure to acceptance.

Accordingly, the findings should not be interpreted as showing that witnessing discrimination causes adolescents to become more discriminatory. They instead suggest that vicarious exposure and acceptance can coexist and that witnessing injustice does not necessarily produce categorical rejection of unequal treatment. Future research should distinguish direct observation from online exposure and examine whether condemnation, peer responses, identification with the target, and perceived social norms determine how vicarious discrimination relates to moral evaluation.

6.2 Perceived discrimination and domain-specific differences

Perceived discrimination produced different findings across the two domains. Perceived ethnic discrimination was not significantly associated with acceptance of ethnic discrimination. In contrast, greater perceived religious discrimination was associated with lower odds of accepting religious discrimination. Neither measure moderated the association between contact and acceptance, meaning that Hypothesis 4 was not supported.

The negative religious association is consistent with an injustice-sensitivity interpretation. Adolescents who believe that they have personally received unfair treatment because of religion may be more attentive to the harm or illegitimacy of similar treatment. Personal experience may make religious discrimination less abstract and could increase recognition of its consequences. Religious identity can be socially visible through clothing, practices, affiliation, or participation, making discriminatory treatment particularly salient in school and peer settings. Research in Belgium suggests that Muslim students may experience religion as a source of belonging and moral orientation while simultaneously encountering exclusion connected to visible religiosity (Van Praag et al., 2016).

This interpretation remains tentative because neither moral sensitivity nor religious identification was measured. It is also possible that adolescents who already strongly reject religious discrimination are more likely to recognize and label ambiguous treatment as religiously discriminatory. The cross-sectional relationship may therefore reflect differences in attribution as well as differences in experience.

The absence of a comparable ethnic association suggests that the consequences of perceived discrimination should not be assumed to operate identically across social categories. Ethnic discrimination may be interpreted through multiple overlapping markers,

including nationality, language, skin colour, ancestry, or migrant background. This could make individual experiences more heterogeneous than religious discrimination as captured by the survey item. The null result may also reflect competing processes. Perceived ethnic discrimination could increase sensitivity to injustice for some adolescents while increasing distrust, disengagement, or defensive group boundaries for others, producing no clear average association.

This complexity is compatible with previous identity research. Discrimination has sometimes been associated with greater centrality of the minority identity, while other studies have found disidentification from the host national community. Jasinskaja-Lahti et al. (2009), for instance, found that discrimination predicted national disidentification among immigrants in Finland, whereas stronger ethnic identification itself did not explain hostility toward the national majority. Ramos et al. (2012) found that discrimination predicted greater identity centrality but not every component of identification. These studies concern different populations and outcomes, but they demonstrate that discrimination does not produce one uniform identity response.

The different ethnic and religious findings support the current study's decision to analyse the domains separately. Had perceived ethnic and religious discrimination been combined into one measure, the negative religious association and null ethnic association could have obscured each other. The domain-specific approach therefore reveals that apparently similar experiences may have different relationships with normative acceptance.

6.3 Migrant background and the absence of group differences

Migrant background was not directly associated with acceptance in either domain, and it did not moderate the relationship between contact and acceptance. Hypothesis 2 was therefore not supported. Adolescents classified as having a migrant background were neither clearly more nor less likely than majority-background adolescents to accept ethnic or religious discrimination once the other predictors and controls were included. The negative contact association also did not differ significantly between the two groups.

The absence of moderation contrasts with the expectation that contact would be less strongly associated with acceptance among migrant-background adolescents because minority-group contact may occur under unequal or discriminatory conditions. However,

some previous adolescent research has also found broadly similar contact patterns across majority and minority groups. Beer ten Berge et al. (2017), for example, found that increases in broader outgroup friendship networks were associated with lower prejudice and that gaining outgroup “foes” was associated with higher prejudice. The effects did not differ substantially between majority- and minority-group pupils.

The present result may indicate that frequent interaction is related to discrimination rejection across group positions. Contact can provide familiarity, information, and opportunities for cross-group relationships for both majority and minority adolescents, even though the meanings and conditions of that contact may differ.

At the same time, the null result should not be interpreted as evidence that adolescents with and without a migrant background occupy equivalent social positions or experience contact in the same way. The migrant-background variable was necessarily broad. It combined adolescents born outside Belgium, Belgian-born children of immigrants, adolescents with one foreign-born parent, and adolescents with two foreign-born parents. It also combined European and non-European origins, different migration histories, ethnic identities, socioeconomic positions, and religious affiliations. Some classified as migrant-background adolescents may be socially perceived as members of the Belgian majority, while others may regularly experience racialization or exclusion.

This heterogeneity may conceal meaningful differences between specific groups. It may also explain why migrant background was not directly associated with acceptance. Group membership alone does not determine normative attitudes, and a binary migration measure cannot capture identification, experiences of marginalization, perceived group status, or political and family socialization. Future studies should distinguish migrant generation, region of origin, ethnic self-identification, religion, and perceived minority status where sample size permits.

6.4 The interaction between ethnic contact and vicarious discrimination

Evidence that vicarious discrimination moderated the contact association appeared only in the ethnic domain. Among adolescents who did not report vicarious discrimination, greater ethnic contact was strongly associated with lower acceptance of ethnic discrimination. Among those who reported vicarious discrimination, the association was weaker and

statistically nonsignificant. The interaction was also reproduced in the larger sample, although its confidence interval was very close to one. No corresponding interaction appeared for religious discrimination.

One possible interpretation is that vicarious discrimination may undermine the extent to which everyday contact is associated with more egalitarian judgments. Positive or frequent interaction may challenge stereotypes at the interpersonal level, while witnessing discrimination may simultaneously signal that ethnic boundaries and exclusion remain socially meaningful. In this situation, adolescents may view their own interethnic contacts as exceptions rather than as evidence that discrimination against the wider group is unacceptable.

This interpretation resembles findings from Tropp et al. (2016), who showed that perceived discrimination altered the relevance of school and peer norms for interethnic experiences. Among minority students perceiving greater discrimination, official school support for cross-group relations appeared less consistently connected to positive experiences, while peer norms sometimes became more important. This suggests that inclusive experiences or messages may lose credibility when they conflict with observed treatment. The present interaction could reflect a similar inconsistency between personal contact and the wider social environment, although this mechanism was not directly tested.

Several reasons require caution. First, the interaction was only slightly below the conventional significance threshold in both the primary and larger samples. Second, several interactions were tested, increasing the possibility of a chance finding. Third, the full model containing all three interactions did not significantly improve fit. Fourth, the effect was absent in the religious domain. Finally, the measure of vicarious discrimination was not domain-specific, so it is unknown whether the observed incidents concerned ethnicity, religion, or another characteristic.

For these reasons, the finding should be presented as tentative rather than conclusive evidence that vicarious discrimination weakens the association between ethnic contact and discrimination rejection. Its value lies in identifying a potentially important direction for future study: contact may need to be considered alongside the wider normative environment in which intergroup encounters occur.

6.5 Secondary gender pattern

Although gender was included as a control rather than a focal predictor, women had significantly lower odds of accepting discrimination in both domains. In the primary models, women had 57% lower odds of accepting ethnic discrimination and 42% lower odds of accepting religious discrimination compared with men.

This pattern is consistent with Kuran and McCaffery's (2008) finding that men expressed greater acceptance across several discrimination scenarios and with the Belgian adolescent profiles identified by De Coninck et al. (2025a), where girls were more frequently classified in the egalitarian profile. However, both studies also found meaningful within-gender heterogeneity. The present result should therefore not be interpreted as showing that all boys are exclusionary or that all girls reject discrimination. Gender may capture differences in socialization, political attitudes, empathy, peer cultures, or willingness to express socially sensitive views, none of which were measured here.

6.6 Practical implications

The findings have several implications for schools and organizations seeking to reduce the normalization of ethnic and religious discrimination. These implications should be interpreted cautiously because the study identifies associations rather than causal effects.

First, the consistent contact findings support creating regular opportunities for adolescents to interact across ethnic and religious boundaries. This does not mean that simply placing diverse students in the same building is sufficient. Research on intergroup contact indicates that interaction tends to be more constructive when it involves cooperation, equal status within the situation, common goals, and clear institutional support (Pettigrew & Tropp, 2006). Cross-group friendships and identification with a shared social community may also be especially valuable (Munniksmma et al., 2015; Pettigrew & Tropp, 2006). In the present study, however, only contact frequency was measured. The results therefore support facilitating interaction while leaving open which specific forms of contact are most effective.

Schools could incorporate sustained cross-group cooperation into ordinary educational practice rather than relying solely on isolated diversity workshops. Relevant approaches may include mixed-group assignments, peer-mentoring initiatives, cooperative extracurricular activities, and structured dialogue in which students work toward shared objectives. These

activities should avoid placing minority students in the position of representing or educating an entire group.

Second, the positive association between vicarious discrimination and acceptance suggests that witnessing discrimination should not be treated as a self-correcting moral lesson. Adolescents may see discriminatory remarks, exclusion, or harassment without observing a clear response from peers, teachers, or institutions. When incidents are ignored or treated as ordinary conflict, they may communicate that discrimination is socially tolerated.

School responses should therefore make the relevant norm explicit. Staff should identify discriminatory conduct accurately, explain why it is unacceptable, and respond consistently rather than leaving observers to infer its meaning. The purpose is not only to support the direct target but also to address the message received by everyone who witnessed the event.

This makes bystander education particularly relevant. Students can be taught how to recognize discrimination, support targeted peers, report incidents safely, and challenge exclusion without placing themselves at unreasonable risk. Such programmes should go beyond encouraging generic kindness. They should help students distinguish discrimination from other interpersonal disagreements and understand how repeated everyday incidents can reinforce wider group hierarchies.

Third, interventions should address both offline and digital environments. The vicarious-discrimination measure included in-person and online exposure, reflecting the fact that adolescents may encounter discrimination at school, within peer groups, or through social media. Schools cannot control all digital content, but they can strengthen students' ability to interpret it critically.

Media-literacy education could examine how discriminatory humour, stereotypes, decontextualized videos, and algorithmically repeated content contribute to perceptions of what is normal. Students should also be encouraged to consider how visible reactions, including approval, silence, or condemnation, shape the social meaning of online incidents. However, because the present analysis did not separate online and in-person exposure, it cannot determine whether one setting is more strongly related to acceptance.

Fourth, the different ethnic and religious findings support domain-sensitive anti-discrimination education. Ethnic and religious discrimination overlap, but they are not interchangeable. Ethnic discrimination may involve nationality, ancestry, language, skin colour, or assumptions about migration. Religious discrimination may involve affiliation,

belief, practice, clothing, or visible participation. Educational interventions should address these specific mechanisms rather than treating all exclusion as a generic diversity problem.

The association between perceived religious discrimination and lower religious-discrimination acceptance may indicate that personally meaningful discussion of religious exclusion can support recognition of injustice. Yet institutions should not assume that direct victimization is educationally beneficial. Discrimination remains harmful, and personal experience should not be required before students understand why unequal treatment is unacceptable. Structured discussion, perspective-taking, and concrete examples can help make the consequences recognizable without exposing students to harm.

Fifth, the findings caution against designing interventions solely around a binary distinction between adolescents with and without a migrant background. The absence of a migrant-background association does not justify ignoring group inequalities. It instead suggests that students within each broad category differ substantially. Programs should be attentive to specific experiences of racialization, religious visibility, migration generation, language, and school belonging rather than presuming that all migrant-background students have the same needs or that all majority-background students occupy the same position.

Finally, schools should evaluate anti-discrimination initiatives rather than assuming that contact, diversity, or awareness automatically produces the intended outcome. Evaluation should examine not only prejudice or liking but also whether students reject discriminatory behaviour, perceive school rules as credible, feel able to intervene, and observe consistent responses from staff. Longitudinal designs would be especially valuable for determining whether changes in contact and school norms are followed by changes in discrimination acceptance.

Taken together, the practical implication is that inclusion requires more than exposure to diversity. Regular cross-group interaction may be beneficial, but its meaning is shaped by the surrounding environment. Schools should combine opportunities for meaningful contact with visible institutional opposition to discrimination, effective responses to witnessed incidents, and education that helps adolescents interpret both ethnic and religious exclusion critically.

6.7 Limitations

Several limitations should be considered when interpreting the findings. These limitations do not negate the consistency of the main results, particularly because the central associations were reproduced in larger-sample and ordinal-outcome sensitivity analyses. They do, however, restrict the conclusions that can be drawn about causality, mechanisms, and generalizability.

First, the study used a cross-sectional observational design. All predictors and outcomes were measured during the same survey period, which prevents conclusions about temporal order or causality. Although greater intergroup contact was associated with lower acceptance of ethnic and religious discrimination, the analysis cannot establish that contact reduced acceptance. Adolescents who already reject discrimination may be more willing to seek, maintain, or report contact with members of other ethnic or religious groups. Both contact and acceptance may also be shaped by unmeasured factors such as parental socialization, political orientation, peer-group norms, personality, school climate, or previous experiences with diversity. Similar concerns apply to perceived and vicarious discrimination. The positive association between vicarious discrimination and acceptance does not demonstrate that witnessing discrimination makes adolescents more accepting of it. Adolescents with more accepting attitudes may instead participate in social or digital environments where discriminatory conduct is more visible. The study therefore identifies patterns of association rather than developmental or causal processes. The thesis design itself appropriately specifies that its models concern associations rather than causal effects.

Second, the study relied entirely on self-reported survey measures. Respondents reported their own intergroup contact, discrimination experiences, and normative attitudes, which may introduce recall error, differences in interpretation, and social-desirability bias. Acceptance of discrimination is a socially sensitive topic, and some adolescents may have understated their acceptance because categorical opposition to discrimination is perceived as the expected response. Conversely, respondents may differ in what they recognize as discriminatory treatment. An adolescent with strong egalitarian commitments may classify an ambiguous event as discrimination, whereas another adolescent may interpret the same event as ordinary conflict or humour. The associations involving perceived discrimination may therefore reflect both differences in treatment and differences in recognition or attribution.

The use of the same response source for predictors and outcomes also raises the possibility of common-method variance. Future conclusions would be stronger if adolescents' reports were combined with peer nominations, teacher reports, behavioural tasks, observations, or administrative information about school environments.

Third, the principal constructs were measured using single survey items or brief indicators, which limits measurement precision. Ethnic and religious contact were measured in terms of the frequency of interaction during the previous month. The measures did not distinguish positive from negative contact, voluntary from involuntary interaction, close friendship from superficial encounter, or contact characterized by cooperation and equal status from contact occurring under conditions of conflict or inequality. Yet contact research indicates that the quality and social conditions of interaction are theoretically important. A student who briefly encounters outgroup members every day may receive the same frequency score as a student who has meaningful and supportive cross-group friendships, despite the likely differences between these experiences.

The perceived-discrimination measures similarly captured broad domain-specific perceptions without identifying the perpetrator, setting, severity, frequency, or precise form of treatment. Discrimination by a close peer, teacher, institution, stranger, or online user may have different implications for adolescents' normative judgments. Multi-item measures would make it possible to assess the reliability of the constructs and distinguish everyday, institutional, overt, and subtle discrimination.

Fourth, the construction of vicarious discrimination was necessarily broad. The variable combined two questions concerning whether respondents had witnessed or encountered discrimination against others in person or online. Respondents were classified as exposed when they answered yes to either question. This operationalization does not distinguish between offline and digital exposure, isolated and repeated incidents, or mild and severe events. It also does not establish who was targeted, which social characteristic motivated the incident, or whether the target was a friend, relative, ingroup member, outgroup member, or stranger.

This is particularly important because the same vicarious-discrimination indicator was used in both the ethnic and religious models. A respondent may have witnessed discrimination based on gender, sexuality, disability, appearance, or another characteristic rather than ethnicity or religion. The measure was therefore not matched to the domain-specific outcomes in the same way as the contact and perceived-discrimination measures.

The variable also contained no information about how the observer or surrounding people responded. Witnessing an incident that is immediately condemned by teachers and peers may communicate a very different social norm from witnessing the same incident without intervention. Research on vicarious discrimination further indicates that its consequences may depend on the observer's relationship with the target, making undifferentiated exposure measures incomplete (Louie & Upenieks, 2022). The unexpected positive association between vicarious discrimination and acceptance should therefore be interpreted as a relationship involving broad exposure, rather than evidence concerning a specific process of normalization.

Fifth, the primary dependent variables were dichotomized to distinguish complete rejection from any level of acceptance. This threshold is substantively meaningful because it captures whether respondents departed at all from categorical rejection. However, it combines adolescents who selected a value only slightly above complete rejection with those expressing much stronger acceptance. Dichotomization consequently reduces variation and may conceal differences in the intensity of attitudes.

This limitation was addressed through supplementary cumulative-link models using the original seven-category outcomes. The ordinal analyses reproduced the main patterns: contact remained negatively associated with acceptance, vicarious discrimination remained positively associated with acceptance, perceived religious discrimination remained negatively associated with religious acceptance, and migrant background remained nonsignificant. The robustness of these findings reduces concern that the substantive conclusions were created by the binary threshold, but the ordinal measures should still be retained as an important supplementary analysis.

Sixth, the measure of migrant background was broad and internally heterogeneous. Respondents were classified as having a migrant background when they or at least one parent had been born outside Belgium. This grouped adolescents born abroad, Belgian-born children of immigrants, adolescents with one foreign-born parent, and adolescents with two foreign-born parents. It also combined EU and non-EU origins and did not distinguish ethnicity, religion, migration history, language, racialization, citizenship, or strength of identification with Belgium or a heritage group.

The absence of a significant migrant-background association therefore does not demonstrate that all migrant-background adolescents hold similar attitudes or that group position is unimportant. Meaningful differences between particular communities may have been obscured by the binary category. Some adolescents categorized as having a migrant

background may be socially perceived as members of the majority, while others may experience frequent ethnic or religious exclusion. Conversely, majority-background adolescents may differ substantially in family ideology, religious affiliation, school environment, and exposure to diversity.

Seventh, the primary analysis used complete-case estimation. The full dataset contained 762 respondents, but the controlled models included 517 respondents with complete information on all required variables. Missingness was particularly high for mother's education, which was unavailable for 146 respondents, while migrant background and vicarious discrimination were also missing for notable numbers of cases. Complete-case analysis may produce selection bias if excluded respondents differed systematically from those retained.

The analysis addressed the most important source of case loss by estimating models without mother's education in a larger sample of 601 respondents. The substantive pattern of results remained similar. Mother's education also improved the fit of the ethnic model, providing a reason for retaining it in the primary specification. Nevertheless, the sensitivity analysis does not eliminate the possibility of selection related to other missing variables. Methods such as multiple imputation could be considered in future analyses, provided that the required assumptions and auxiliary variables can be defended.

Eighth, the sampling procedure limits population generalizability. Schools were selected through quota sampling based on province and educational orientation, after which participating schools selected classes. The sample therefore was not a probability sample of all adolescents in Belgium. It covered pupils in the third cycle of Dutch-language secondary education, mainly between 16 and 18 years old, and cannot automatically be generalized to younger adolescents, French-language schools, adolescents outside formal education, or youth in other national contexts. School participation and class selection may also have produced selection if schools or teachers particularly willing to participate differed in their approaches to diversity and discrimination.

Ninth, the data had a hierarchical structure, with pupils sampled through classes and schools, but school and classroom identifiers were unavailable for the secondary analysis. The regression models could consequently not account for potential dependence among respondents from the same educational setting. The reported standard errors may be too narrow if attitudes and experiences were correlated within schools or classes.

The school-location control distinguished city, suburb, village, and countryside locations, but this captures only a broad geographical context. It does not measure school

composition, classroom diversity, institutional norms, teacher responses to discrimination, or peer climates. Previous research shows that school and peer norms may be associated with interethnic comfort, friendship interest, and contact quality, illustrating the relevance of contextual variation (Tropp et al., 2016).

Finally, the moderation findings require particular caution. Six domain-specific interaction tests were conducted, and only the interaction between ethnic contact and vicarious discrimination reached conventional statistical significance. Its significance was modest in the primary sample, $p = .040$, and remained close to the threshold in the larger sample, $p = .047$, with a confidence interval approaching one. No corresponding interaction appeared in the religious domain, and the model containing all interactions did not provide clear evidence of superior fit. The analysis itself correctly characterizes this result as partial and tentative support requiring replication. Given the lower statistical power of interaction tests and the possibility of chance findings across multiple comparisons, this result should not carry the same evidential weight as the replicated main effects.

6.8 Future research

Future research should first use longitudinal designs to establish temporal order. Following adolescents across several school years would help determine whether increases in intergroup contact precede changes in discrimination acceptance, whether existing attitudes shape subsequent contact choices, and whether the relationship operates reciprocally. Longitudinal models that separate stable differences between adolescents from changes occurring within the same adolescent would be especially valuable. Recent youth contact research has found strong between-person associations but limited evidence that within-person increases in contact consistently generate later improvement, demonstrating why this distinction matters (McKeown et al., 2025).

Experimental and intervention studies could provide stronger causal evidence. For example, schools could evaluate whether structured cooperative contact, cross-group friendship programmes, or classroom discussions of discrimination alter normative judgments compared with appropriate control conditions. Such studies should include follow-up measurements to assess whether changes persist beyond the immediate intervention. Reviews of school-based intergroup programmes recommend random assignment where possible,

appropriate controls, multiple informants, systematic implementation assessment, and integration into continuing educational practice rather than isolated short-term activities (Gönültaş & Mulvey, 2019).

Second, future studies should adopt multidimensional contact measures. In addition to frequency, researchers should distinguish:

- Positive and negative contact
- Friendship and casual interaction
- Voluntary and involuntary contact
- Online and offline contact
- Equal-status and unequal-status encounters
- Cooperative and competitive interaction
- Contact in classrooms, neighbourhoods, extracurricular activities, and digital spaces

Including mediating variables such as intergroup anxiety, empathy, perceived threat, familiarity, shared identity, and stereotype change would help explain why contact is associated with the rejection of discrimination. This is particularly relevant because the present study demonstrates an association with normative acceptance but cannot identify the psychological or social process underlying it.

Third, the finding concerning vicarious discrimination calls for more differentiated observational measures. Future surveys should record:

- The social group targeted
- Whether exposure occurred online or in person
- The frequency, severity, and recency of incidents
- The observer's relationship with the target
- Whether the target was perceived as an ingroup or outgroup member
- The observer's emotional and moral reaction
- Whether peers, teachers, or institutions condemned the incident
- Whether the observer intervened, remained silent, or supported the perpetrator

These distinctions would allow researchers to test competing explanations. Exposure accompanied by condemnation may increase injustice awareness, whereas repeated exposure without sanction may communicate that discrimination is normal or tolerated. Exposure involving a close ingroup member may evoke solidarity or threat differently from exposure involving a stranger. Separating these processes is necessary before normalization can be treated as an explanation for the positive association found here.

An intersectional approach would also be beneficial. Adolescents' experiences may reflect combinations of ethnicity, religion, gender, social class, and educational track rather than one isolated category. Larger samples would permit comparisons among specific groups without collapsing very different social positions into one category.

Fourth, future data collection should preserve school and classroom identifiers and measure contextual features directly. Multilevel models could then distinguish individual-level relationships from differences between classes and schools. Relevant contextual measures include:

- Ethnic and religious school composition
- Educational orientation
- School equality policies
- Teacher and peer norms
- Institutional responses to discriminatory incidents
- Perceived school safety and belonging
- Opportunities for cooperative cross-group interaction
- Online peer climates

This would make it possible to examine whether contact is more strongly associated with rejection of discrimination in schools where inclusive norms are credible and consistently enforced. It would also allow researchers to determine whether the vicarious-discrimination association reflects individual exposure or broader climates in which discriminatory conduct is normalized.

Fifth, replication should involve broader and more representative samples. Probability-based studies covering both Dutch- and French-language education would provide a stronger basis for conclusions about Belgian adolescents. Cross-national comparisons could examine whether the findings depend on differences in migration histories, religious institutions, educational systems, equality policies, and public discourse. Younger samples would also help determine when acceptance of ethnic and religious discrimination begins to differentiate and how it changes across adolescence.

Finally, future studies should test the tentative moderation result through preregistered and adequately powered analyses. Replication should use domain-specific measures of witnessed ethnic and religious discrimination, correct for multiple interaction tests, and estimate predicted probabilities across theoretically meaningful values. Until such evidence is available, the ethnic contact-by-vicarious-discrimination interaction is best understood as a promising hypothesis rather than an established conclusion.

Taken together, these limitations point toward a research agenda that combines longitudinal and multilevel designs with richer measures of contact, identity, school norms, and direct and vicarious discrimination. Such work would help move beyond documenting who accepts discrimination toward explaining how everyday interactions and social environments contribute to the development, normalization, or rejection of unequal treatment.

7 Conclusion

This study examined how intergroup contact, discrimination experiences, and migrant background were associated with adolescents' acceptance of ethnic and religious discrimination in Dutch-language Belgian secondary schools. By analysing ethnic and religious discrimination separately, the study assessed whether the same relationships appeared across two connected but distinct domains. Overall, the findings indicate that adolescents' acceptance of discrimination is most consistently associated with the frequency of their interactions with members of other ethnic and religious groups.

Intergroup contact was the clearest result. More frequent ethnic contact was associated with lower acceptance of ethnic discrimination, while more frequent religious contact was associated with lower acceptance of religious discrimination. These associations were similar in strength and remained present after demographic and school-location controls were included. They were also reproduced when the original seven-category acceptance measures were analysed and when a larger sample excluding mother's education was used. Although the cross-sectional design does not establish that contact causes adolescents to reject discrimination, the consistency of the findings suggests that adolescents who interact more frequently across ethnic and religious boundaries are less likely to regard unequal treatment as acceptable.

Vicarious discrimination produced a different pattern. Adolescents who reported witnessing or encountering discrimination directed at others were more likely to express some acceptance of both ethnic and religious discrimination. This finding is noteworthy because exposure to injustice does not appear to correspond automatically with stronger rejection of discriminatory treatment. However, it remains observational and should not be interpreted as evidence that witnessing discrimination makes adolescents more accepting of it. The measure combined online and in-person exposure and did not identify the target, context, frequency, severity, or type of discrimination observed. The association may reflect normalization,

differences in peer or digital environments, selective exposure, or other unmeasured processes.

Direct perceived discrimination showed no consistent cross-domain relationship. Perceived religious discrimination was associated with lower acceptance of religious discrimination, whereas perceived ethnic discrimination was not significantly associated with ethnic-discrimination acceptance. This difference supports analysing ethnic and religious discrimination separately rather than assuming that they operate through identical processes. Migrant background was not clearly associated with either outcome, although the broad binary measure may have concealed important differences in migration generation, origin, religious affiliation, racialization, and identification.

The evidence that these relationships varied between groups was weak. Migrant background and perceived discrimination did not significantly moderate the association between contact and acceptance. Vicarious discrimination moderated the contact association only in the ethnic domain, where contact was more strongly associated with rejection among adolescents who did not report vicarious discrimination. Because this interaction was modest, close to the significance threshold, and not reproduced in the religious domain, it should be treated as tentative rather than as a firm conclusion.

These conclusions are limited by the cross-sectional design, complete-case analysis, reliance on single-item measures, and the absence of school and classroom identifiers. The study therefore cannot establish causal direction, fully represent the complexity of contact and discrimination experiences, or account for clustering within educational settings. Nevertheless, the findings show that normative acceptance deserves attention as an outcome in its own right. The central conclusion is that intergroup contact is consistently associated with stronger rejection of ethnic and religious discrimination, while exposure to discrimination has more complex and domain-dependent relationships with adolescents' judgments about unequal treatment.

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9 Appendix

9.1 Appendix A: Operationalization and Coding of Study Variables

Appendix A: Operationalization and coding of analysis variables			
Construct	Survey item(s)	Original measure	Analytical coding
Ethnic discrimination acceptance	Q20_1	1-7; higher scores indicate greater acceptance	Primary outcome: 0 = complete rejection (score 1); 1 = any acceptance (scores 2-7). Original 1-7 scale retained for ordinal sensitivity analysis
Religious discrimination acceptance	Q20_4	1-7; higher scores indicate greater acceptance	Primary outcome: 0 = complete rejection (score 1); 1 = any acceptance (scores 2-7). Original 1-7 scale retained for ordinal sensitivity analysis
Ethnic intergroup contact	Q15_1	1-6; higher scores indicate more frequent contact	Continuous; mean-centred within the ethnic analytic sample
Religious intergroup contact	Q15_7	1-6; higher scores indicate more frequent contact	Continuous; mean-centred within the religious analytic sample
Perceived ethnic discrimination	Q16_1	1-5; higher scores indicate greater perceived discrimination	Continuous; mean-centred for interaction models
Perceived religious discrimination	Q16_4	1-5; higher scores indicate greater perceived discrimination	Continuous; mean-centred for interaction models
Vicarious discrimination	Q18 and Q19	Exposure to discrimination directed at others in person or online	0 = no to both items; 1 = yes to either item; insufficient information = missing
Migrant background	Q42 and Q43	Respondent and parental country of birth	0 = respondent born in Belgium and both parents born in Belgium; 1 = respondent or at least one parent born outside Belgium
Gender	Q32	Man, woman, non-binary, other, prefer not to say	Man = reference; Woman; Other/undisclosed
Age	Q33	15, 16, 17, 18, 19, other	Continuous age in years; 'Other' treated as missing
Mother's education	Q37	None, primary, secondary, college, university, PhD, don't know, other	Secondary education = reference; Low education = none/primary; Higher education = college/university/PhD; unknown/other coded separately before complete-case selection
School environment	Q38	Countryside, village, suburb, city	City = reference; Suburb; Village; Countryside

Note. Higher scores on the original acceptance, intergroup-contact, and perceived-discrimination measures indicate greater acceptance, more frequent contact, and greater perceived discrimination, respectively. Reference categories for categorical predictors are shown in the analytical-coding column. The original seven-category acceptance measures were retained for ordinal sensitivity analyses.

9.2 Appendix B: Analytic Sample Construction and Missing Data

Appendix B1: Sample construction

Sample	N
Full dataset	762
Ethnic model core variables complete	617
Religious model core variables complete	617
Ethnic final analytic sample including controls	517
Religious final analytic sample including controls	517

Note. The full dataset contained 762 respondents. The primary analyses used complete-case estimation. The final analytic samples included respondents with complete information on the outcome, focal predictors, migrant

background, vicarious discrimination, and all control variables. The same 517 respondents were used across the primary models within each domain.

Appendix B2: Missing data for analysis variables

Variable	Missing_N	Missing_Percent
mother_education_control	146	19.2
migrant_background	93	12.2
vicarious_discrimination	74	9.7
gender_control	46	6.0
age_control	41	5.4
school_environment_control	30	3.9
accept_ethnic_any	13	1.7
accept_ethnic	13	1.7
accept_religious_any	13	1.7
accept_religious	13	1.7
perceived_ethnic	11	1.4
perceived_religious	11	1.4
contact_ethnic	7	0.9
contact_religious	7	0.9

Note. Missing counts and percentages are calculated relative to the full sample of $N = 762$. Mother's education had the highest level of missingness (19.2%). Primary models were estimated using complete cases across all variables included in the model specification.

9.3 Appendix C: Diagnostic Checks for the Primary Regression Models

Appendix C1: Generalized variance inflation factors for the primary controlled models

Domain	Predictor	GVIF	df	Adjusted GVIF
Ethnic discrimination	Ethnic intergroup contact	1.09	1	1.04
Ethnic discrimination	Perceived ethnic discrimination	1.50	1	1.23
Ethnic discrimination	Migrant background	1.46	1	1.21
Ethnic discrimination	Vicarious discrimination	1.04	1	1.02
Ethnic discrimination	Gender	1.06	1	1.03
Ethnic discrimination	Age	1.08	1	1.04
Ethnic discrimination	Mother's education	1.26	2	1.06
Ethnic discrimination	School environment	1.12	3	1.02
Religious discrimination	Religious intergroup contact	1.05	1	1.02
Religious discrimination	Perceived religious discrimination	1.21	1	1.10
Religious discrimination	Migrant background	1.27	1	1.13
Religious discrimination	Vicarious discrimination	1.05	1	1.02
Religious discrimination	Gender	1.05	1	1.02
Religious discrimination	Age	1.08	1	1.04
Religious discrimination	Mother's education	1.21	2	1.05
Religious discrimination	School environment	1.09	3	1.01

Note. GVIF = generalized variance inflation factor. For predictors with more than one degree of freedom, the adjusted measure $GVIF^{1/(2 \times df)}$ is reported to facilitate comparison across predictors. Adjusted GVIF values ranged from 1.01 to 1.23, indicating no evidence of problematic multicollinearity.

Appendix C2: Tests of linearity in the logit for continuous predictors

Domain	LR chi-square	df	p-value
Ethnic discrimination	4.40	3	.221
Religious discrimination	3.91	3	.272

Note. Likelihood-ratio tests compare the primary linear specifications with models additionally containing quadratic terms for intergroup contact, perceived discrimination, and age. Non-significant tests indicate that the quadratic specification did not provide significantly better fit than the linear specification.

Appendix C3: Influence diagnostics for the primary and retained interaction models

Model	N	Maximum Cook's distance	Cases above 4/N	Cases above Cook's D = 1	Maximum leverage	Cases above $2p/N$	Cases with absolute residual above 3
Ethnic primary model	517	0.019	18	0	0.095	31	0
Religious primary model	517	0.021	17	0	0.081	34	0
Ethnic interaction model	517	0.021	18	0	0.097	33	0

Note. Cook's distance and leverage thresholds of $4/N$ and $2p/N$, respectively, were used as preliminary screening criteria, where N is the model sample size and p is the number of estimated parameters. No observations exceeded Cook's $D = 1$ or an absolute standardized deviance residual of 3. No observations were excluded on the basis of these diagnostics.

9.4 Appendix D: Sensitivity and Ordinal-Model Specification Checks

Appendix D1: Likelihood-ratio tests for the contribution of mother's education

Domain	LR	chi-square	df	p-value
Ethnic	9.967	2	0.007	
Religious	3.648	2	0.161	

Note. Likelihood-ratio tests compare otherwise identical models estimated on the same $N = 517$ complete-case samples, with and without mother's education. A significant test indicates that including mother's education significantly improves model fit.

Appendix D2a: Nominal-effects tests of the proportional-odds assumption

Domain	Predictor	LR	chi-square	df	p-value
Ethnic discrimination	Intergroup contact	8.81	5	.117	
Ethnic discrimination	Perceived discrimination	5.32	5	.378	
Ethnic discrimination	Migrant background	3.64	5	.603	
Ethnic discrimination	Vicarious discrimination	4.83	5	.437	
Ethnic discrimination	Gender	18.75	5	.002	
Ethnic discrimination	Mother's education	4.27	10	.934	
Religious discrimination	Intergroup contact	8.61	5	.126	
Religious discrimination	Vicarious discrimination	4.25	5	.514	
Religious discrimination	Gender	8.20	5	.145	
Religious discrimination	Mother's education	9.65	10	.472	

Note. Nominal-effects tests assess whether allowing a predictor's effect to vary across cumulative outcome thresholds significantly improves model fit. A significant result indicates evidence against the proportional-odds assumption for that predictor. Tests that could not be estimated reliably are omitted. Gender showed evidence of non-proportionality in the ethnic model and was therefore allowed to vary across thresholds in the partial proportional-odds specification.

Appendix D2b: Comparison of proportional and partial
proportional-odds ethnic models

Model	Log likelihood	AIC
Proportional-odds model	-573.82	1181.64
Partial proportional-odds model	-564.45	1172.89

Note. Lower AIC values indicate better relative model fit. The partial proportional-odds model allows the effect of gender to vary across outcome thresholds while retaining proportional effects for the remaining predictors. The partial model provided significantly better fit than the standard proportional-odds model, LR $\chi^2(5) = 18.75$, $p = .002$.